Anthony Wang

Mechanical Engineering Portfolio

Intro

Anthony Wang received his M.S. in Mechanical Engineering and specialized in Robotics from New York University in May 2020. He earned his B.S.M.E from Boston University in 2018 with a manufacturing concentration. Recent accomplishments include winning the Mohamed Bin Zayed International Robotics Competition (MBZIRC), and the NYU InnoVention Startup Competition.

Currently, Anthony researches at the NYU Agile Robotics and Perception Lab and works at NYU MakerSpace where he teaches students on CAD/CAM and prototyping techniques.

Anthony has a strong experience in product development, mechanical system design, and robotics. He previously worked in the Robotics, Consulting, and Manufacturing industries.

This portfolio summarizes the projects he did in school, during his work, as well as some crafts he did during his spare time.

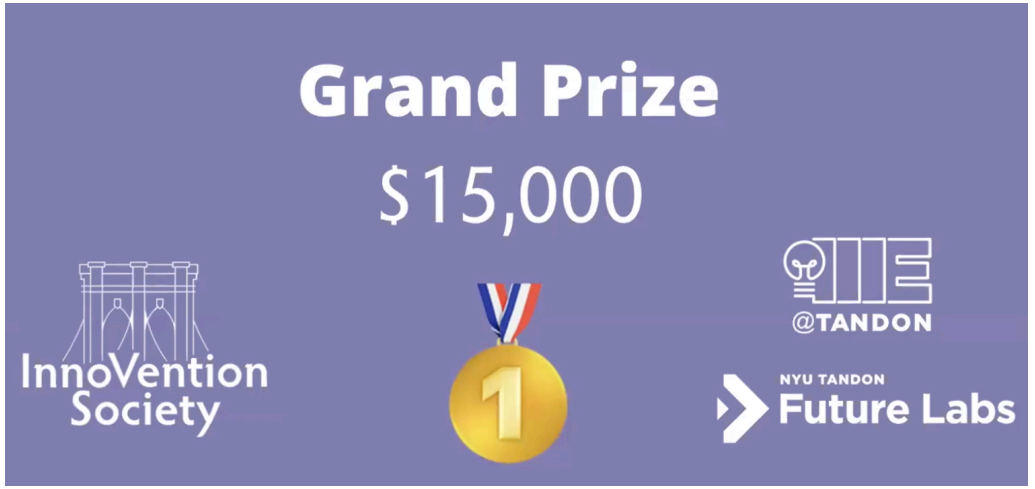
Latest Accomplishments

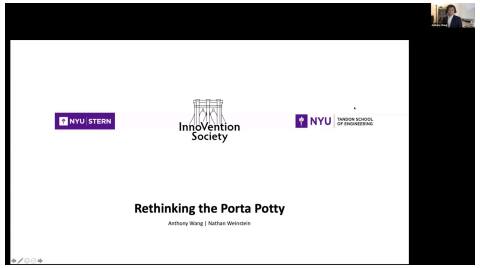
- In February 2020, Anthony attended the Mohamed Bin Zayed International Robotics Challenge (MBZIRC) in Abu Dhabi, won two individual awards, and took home the Grand Challenge Award.
- The team consists of hand-picked robotics elites from NYU, Czech Technical University, and U Penn.
- Anthony worked on the trajectory planning and control of a 7DOF mobile manipulator, hardware design, and sensor integrations.
- The challenge he participated in involved putting out fire in a high rise building as well as a pick and place challenge for wall construction. Both had to be done autonomously.

https://engineering.nyu.edu/news/nyu-tandon-students-triumph-global-robotics-competition

Latest Accomplishments

- Anthony participated in the NYU InnoVention Competition, a semester long tech startup challenge, where 16 NYU teams build and pitch a commercially viable product to compete for a $50,000 prize pool.
- Anthony's Product – Rethinking the Porta Potty aims to solve the space utilization issue in the portable sanitation industry.
- Anthony's pitch won the grand prize and voted by audience for "most likely to invest" venture.
- For information on the product in the page 5.





Competition held virtually due to COVID-19

[Winning Pitch Video]

https://engineering.nyu.edu/news/innoventions-first-ever-virtual-finals

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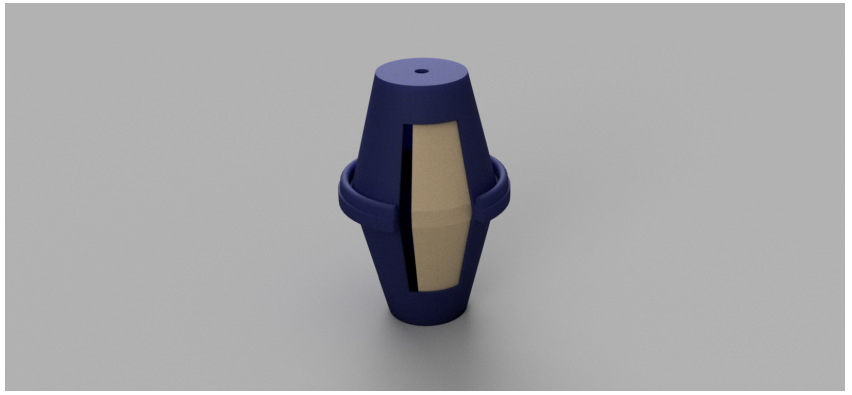
Rethinking Porta Potty

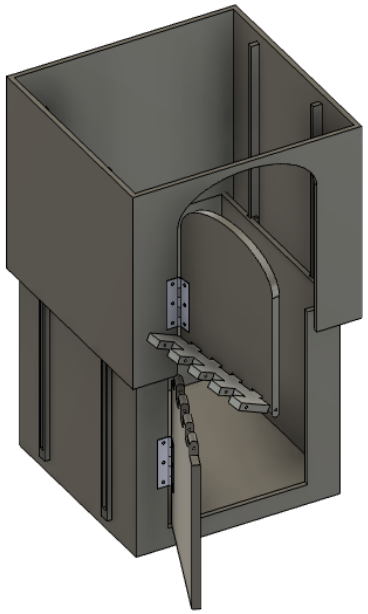
- Project Description:
  - This project began as a part of NYU Prototype Fund in 2019, a semester long program from NYU Entrepreneurial Institute to prototype any approved submissions with a $500 grant and mentoring services.
  - After the first successful prototype build, our product was selected to advance to the InnoVention Competition, a tech startup challenge consists of teams from NYU and NYU Abu Dhabi.
- Motivation and Summary of Project:
  - Porta Potty's consists of 90% empty space, so the stacking of units for transportation and storage is highly inefficient.
  - Pricing of one porta potty unit ranges from $750 to $3000, highly overpriced for a piece of molded plastic.
  - Our goal is to create a porta potty unit that can reduce in size when being stacked, without sacrificing the structural integrity.
  - The unit will be cheap, easy to operate, and reduces the carbon footprint through space savings and fuel efficiency.

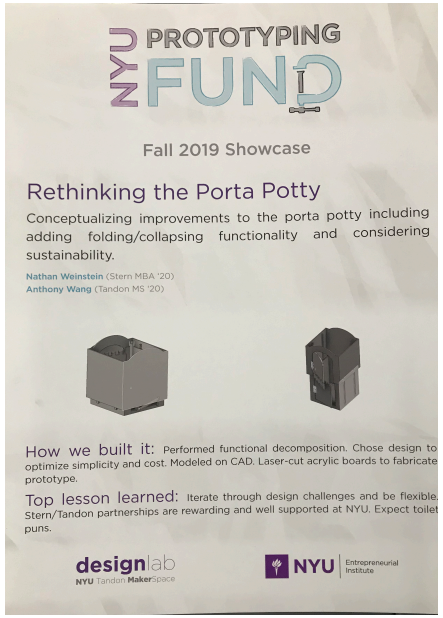


Prototype Version 1:

Achieve 50% space reduction through collapsing features, the space that would originally fit one unit can now fit two units
Prototyped through laser-cut acrylic sheets

Problems: Complex design due to moving mechanisms, only 50% space reduction, we can do better!



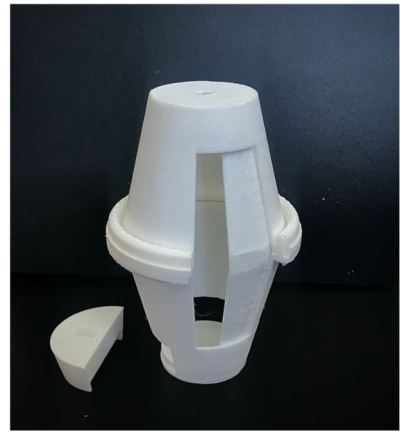


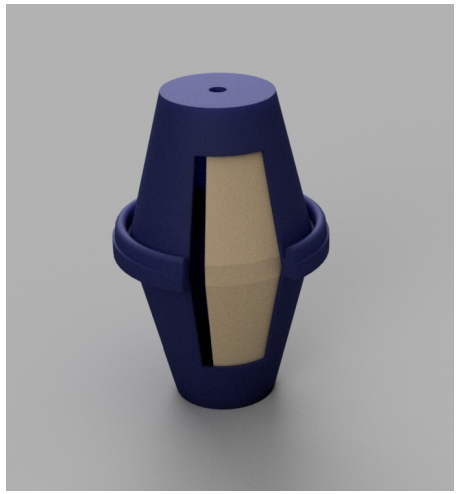
Project Showcase | CAD Model | Showcase Flyer

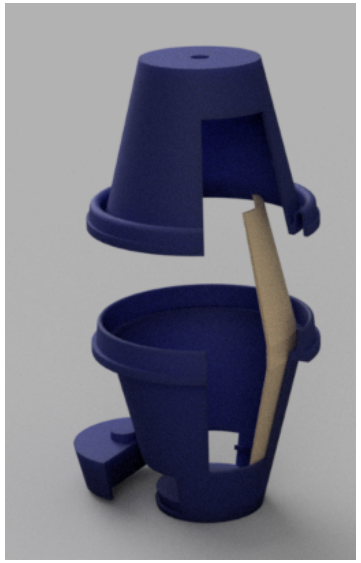
Media Coverage: https://engineering.nyu.edu/news/fall-prototyping-fund-showcase-displays-diversity-and-creativity

Prototype Version 2 (InnoVention Competition)

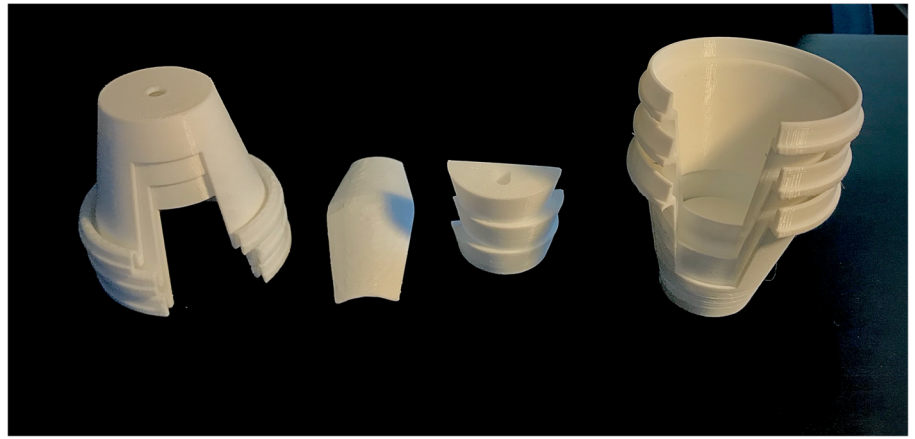
- Stackable like solo cups, massive space reduction for transport and storage
- Infinite Space Savings: The more you stack, the more space you save
- Circular design: No Stress Concentration, sleek and modern design
- No moving mechanisms for easy and fast assembly process







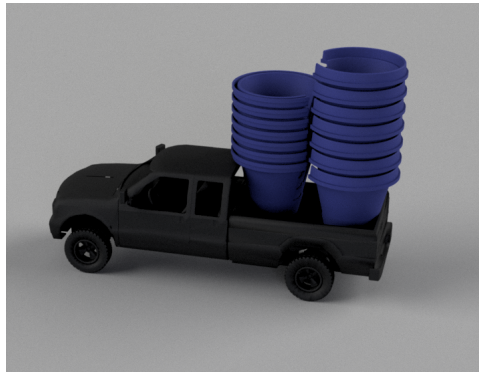
Rendered Images of Final Product



Fully Functional 3D Printed Prototypes

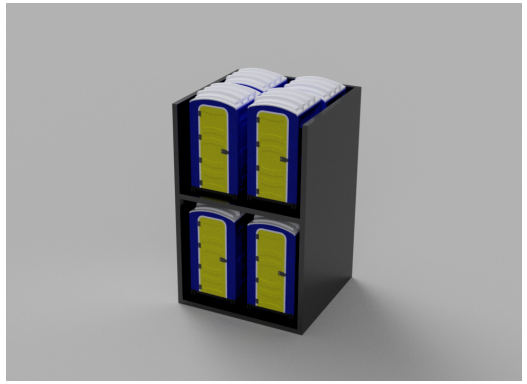
Resulting Savings from the New Design

Transport Savings



$24 K
fuel savings annually[1]

6.6 Ton CO_2 emission reduction per vehicle



Storage Savings

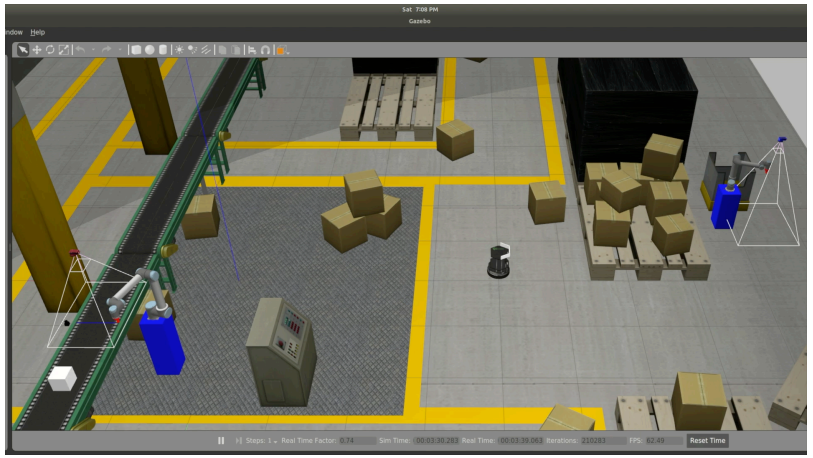
Same Warehouse
8 Units vs 24 Units

+ 62% space efficiency

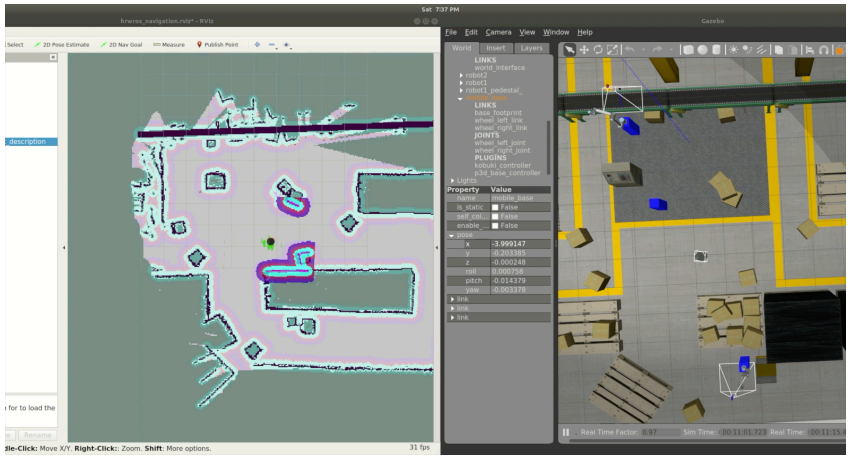
1. Appendix 7: Example 2 on Fuel Savings for Fleet Operators, RTPP

Factory Automation Simulation

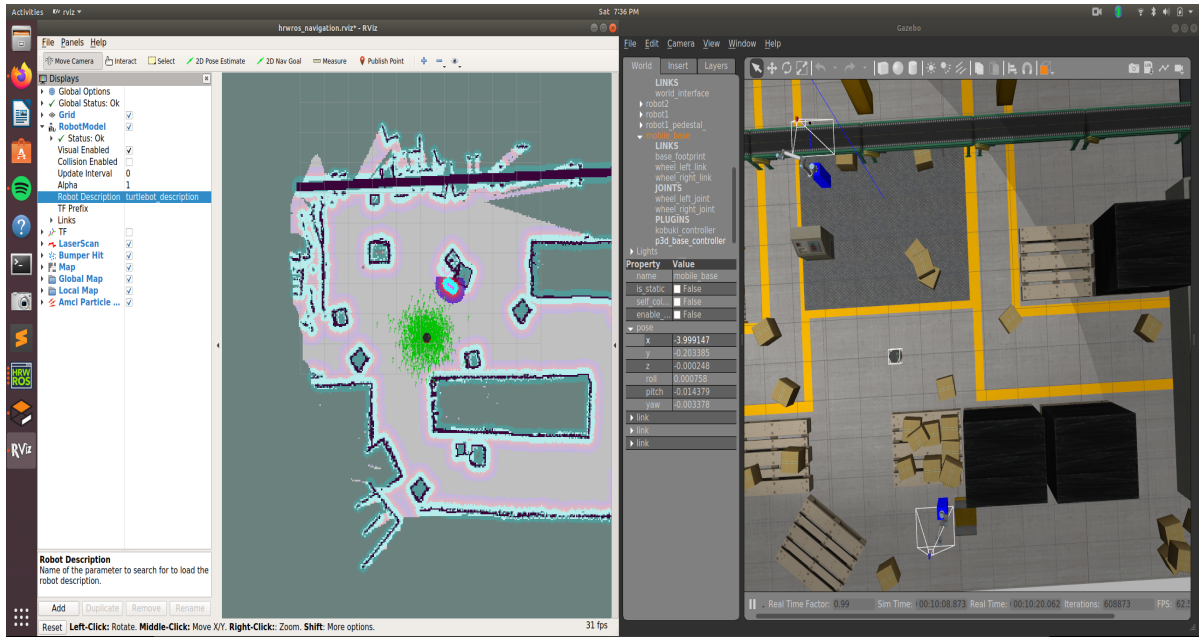
- Gazebo Environment adapted from the Gilbreth pick and place industrial application developed for ARIAC
- Simulated warehouse automation scenario with turtle bot, UR manipulators to collaboratively complete a pick and place task (see video)
- Different modules including:
    1. Configuring Moveit! planning framework for the two UR robot manipulation
    2. Flexbe for state machine and robot behavior design
    3. Gazebo Plugins for gripper, camera simulation
    4. Adaptive Monte Carlos Localization for SLAM
    5. Logical camera for computer vision
- All nodes written in Python and integrated together in ROS

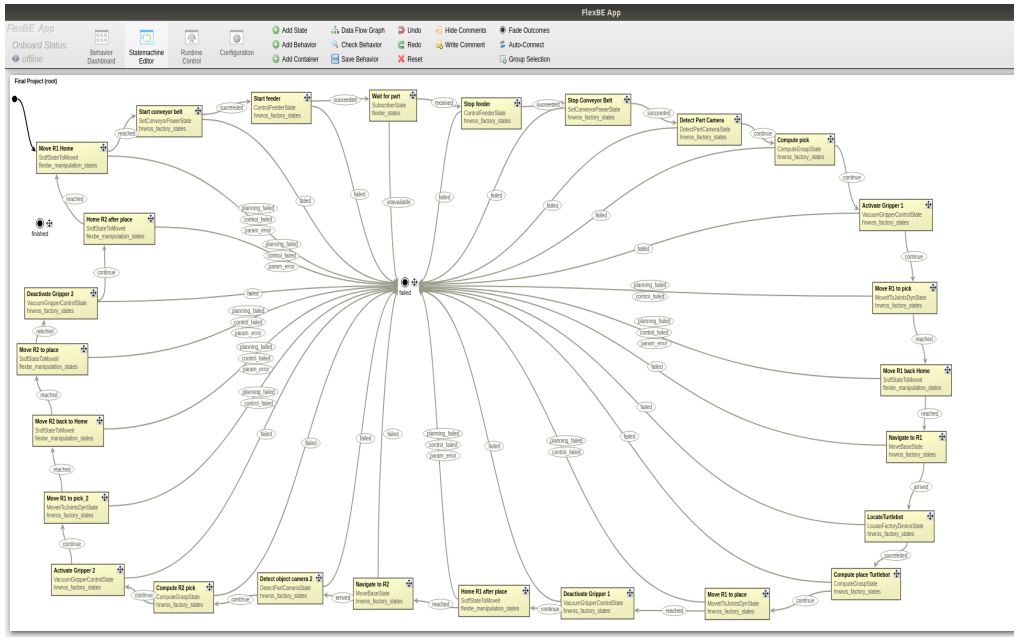


Link to Simulation Link to SLAM demo



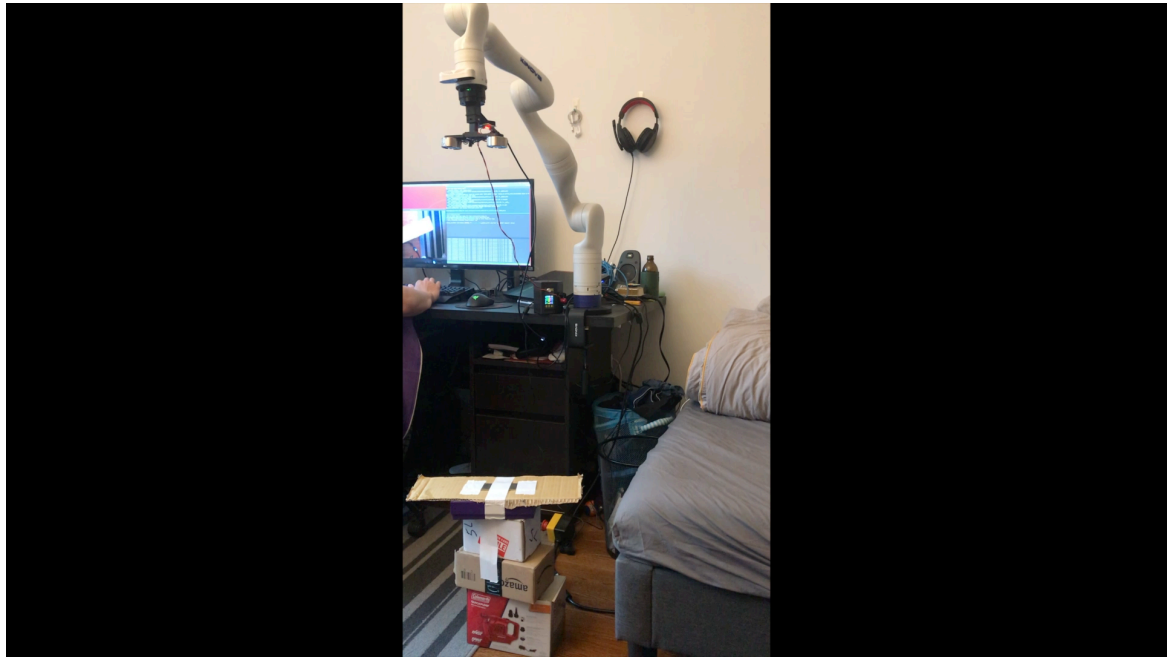
SLAM Module and State Machine Design





Autonomous Pick and Place with Kinova Arm

- Part of Anthony's M.S. Project, extension of MBZIRC brick pick and place challenge
- Using Vision System and image processing technique to identify to 3D Pose of object
- Using Moveit! to compute and execute trajectory
- Using magnetic actuator to grip detected object
- Using SMACH to design robot behavior

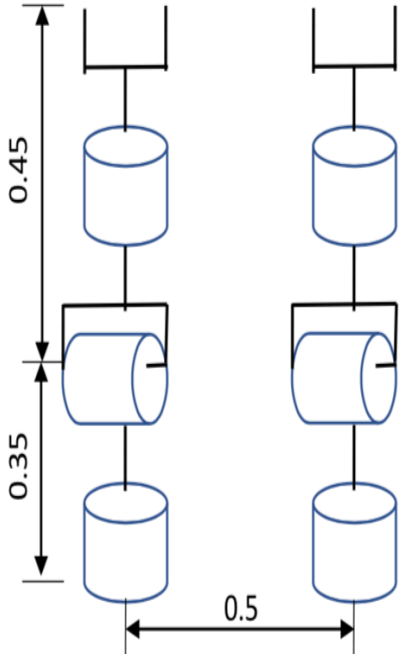


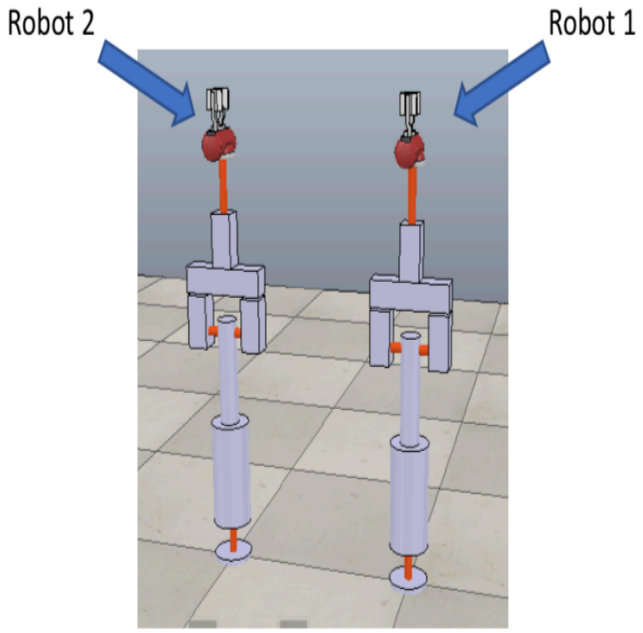
Link to Video

Coordinated Motion of Two Robot Arms

- Project Description:

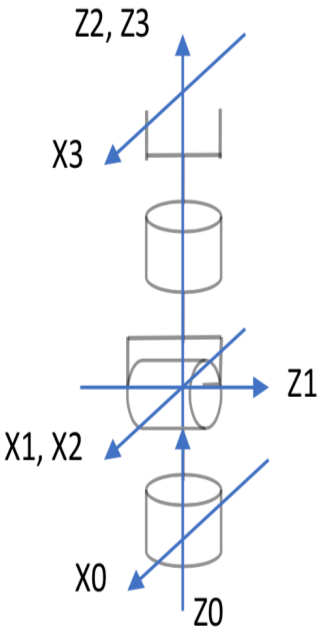
  - Plan and simulate a pair of spherical wrist manipulators consisting three revolute joints
  - Virtual Robot Experimental Platform (VREP) for simulation and Matlab for calculations and remote API
  - Analyze the fundamental motions of the manipulator duo to perform tasks in both “follow mode” and “mirror mode”
  - Implement collision-proof and path planning Algorithm
  - Look into potential industrial applications and simulate beer pouring motion.





Project Planning

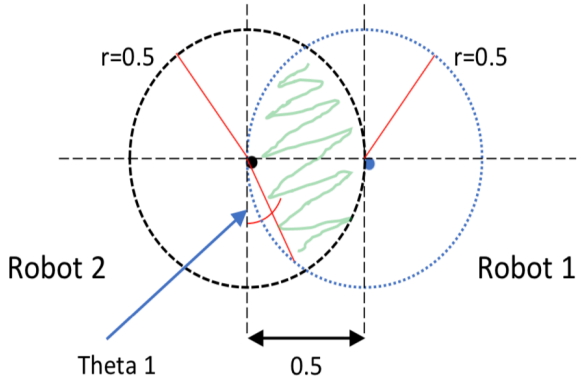
- Denavit - Hartenberg Coordinates and Parameters



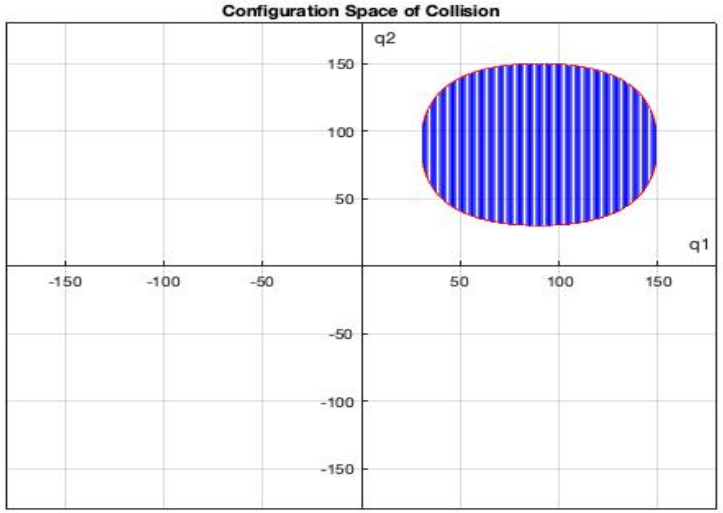
link	a_i	α_i	d_i	θ_i
1	0	$-90°$	d1	θ_1
2	0	$90°$	0	θ_2
3	0	$0°$	d3	θ_3

Figure 3

- Denavit – Calculating Collision Conditions and establishing in configuration space



r = d3 + height of gripper
= 0.35 + 0.15
$\theta_1 = \sin^{-1}\left(\frac{0.25}{0.5}\right)$ in this case.



Results

- Mirror Mode with Collision Warning: Left Robot Mirrors the Action of Right Robot, when the collision algorithm detects a crash between two robots, the action is immediately ceased and warning is displayed.

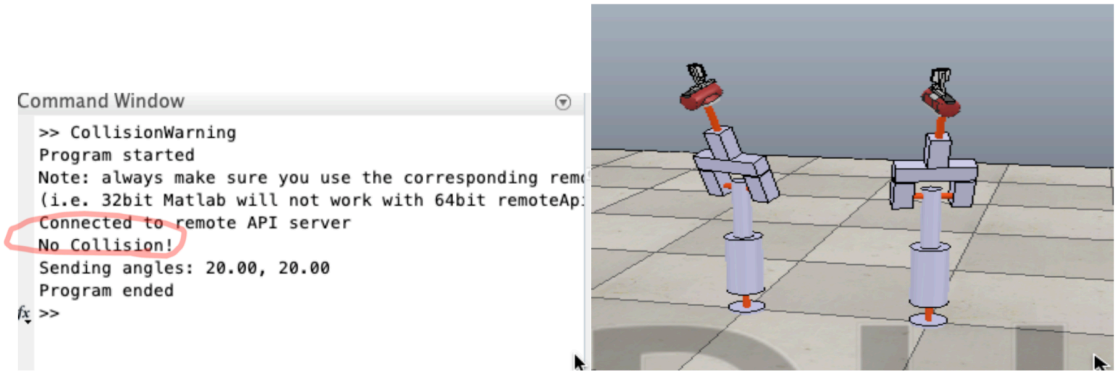


Figure 8, No Collision

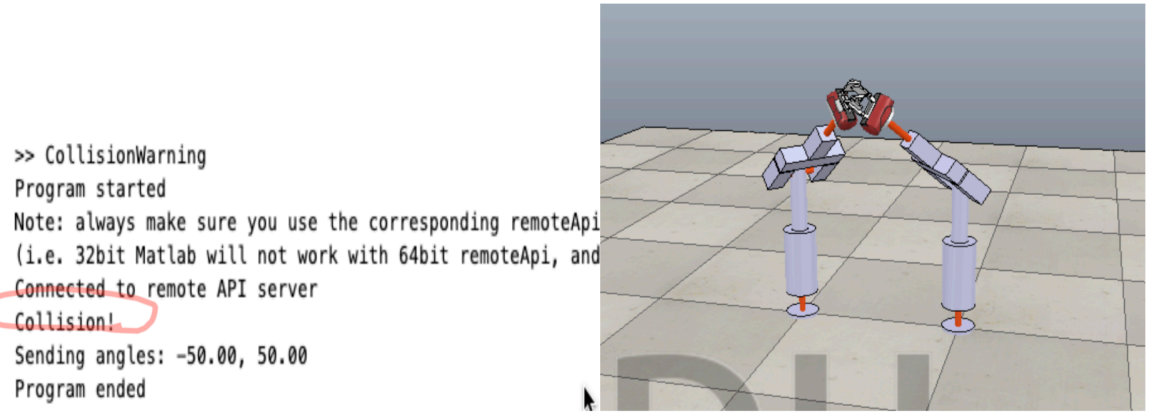


Figure 9, Collision Occurs

- Demo of beer pouring application

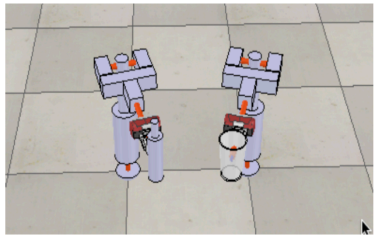


Figure 15

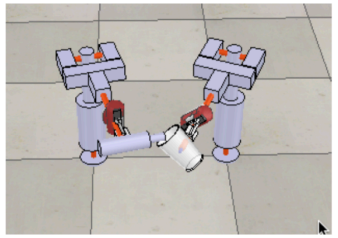


Figure 16

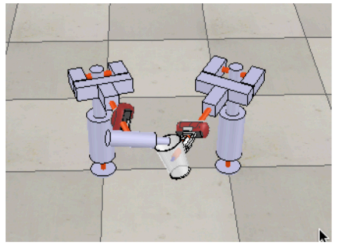


Figure 17

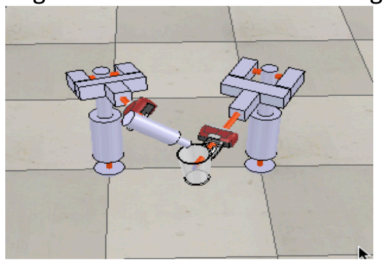


Figure 18

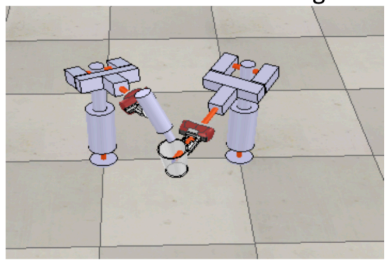
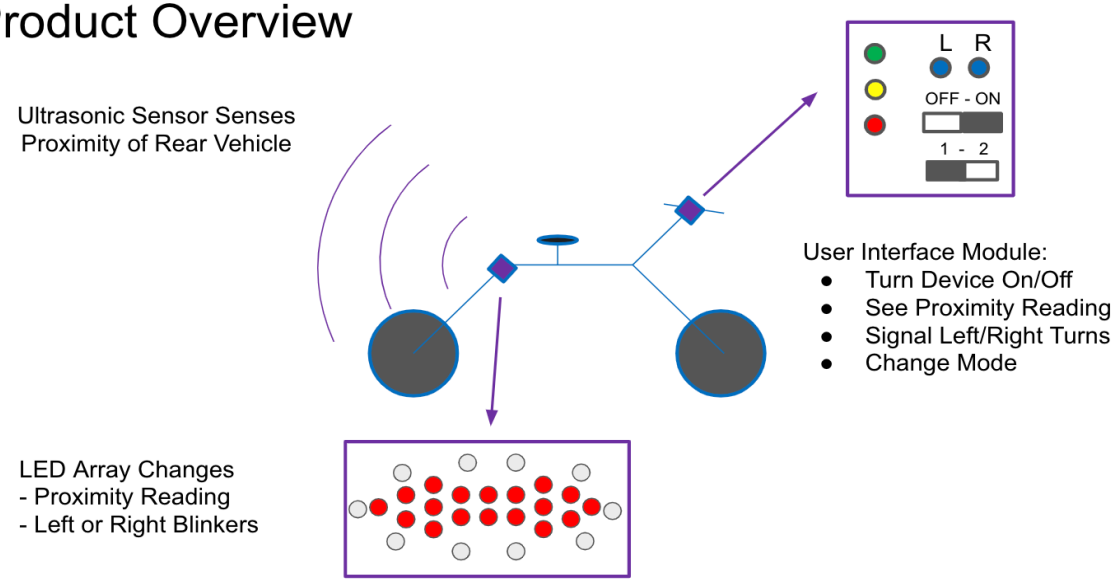


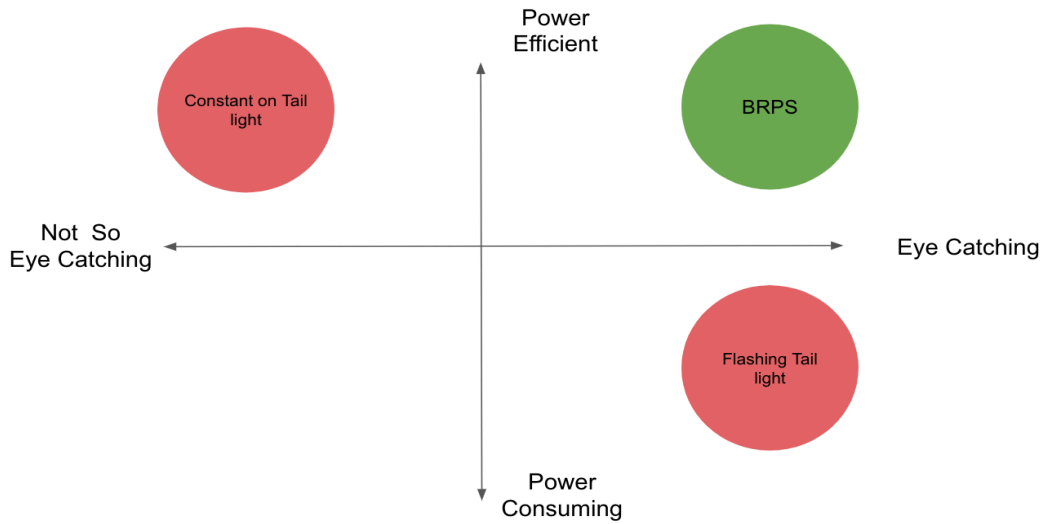
Figure 19

Bike Rear Protection System (BRPS)

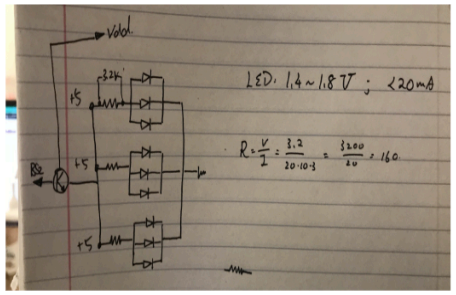
- Problem Statement: Traditional LED taillights are not smart. They are either constantly on (Power Efficient but not alerting) or constantly flashing (Alerting but power consuming). There is not common ground in between the two modes without user manually switching
- Project Description:
  - Build a system to prevent cyclists from rear end collision by more effectively alerting rear end oncoming drivers
  - LED Display flashes different patterns based on the proximity of vehicle behind
  - Generate left and right turn signals upon user inputting direction on UI
  - Provide cyclist feedback of rear proximity with LED indicators on UI

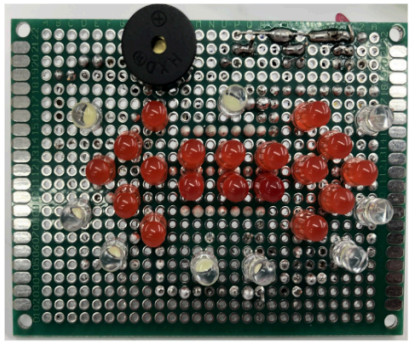
Product Overview

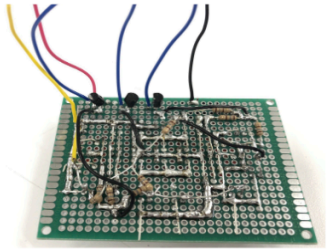




Prototype Build - LED Driver

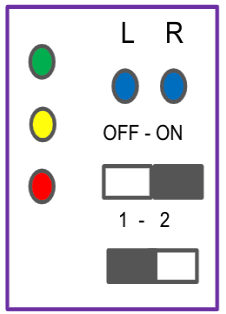




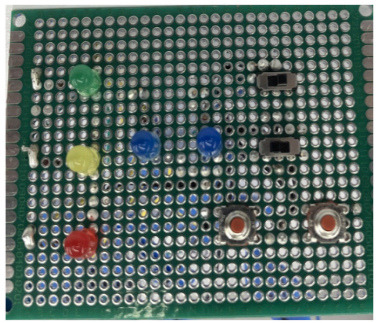


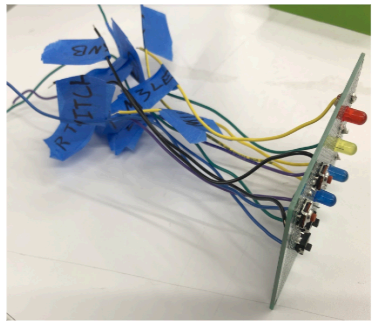
* Separating actuator circuit using NPN transistors

Prototype Build - User Interface



UI Design Concept



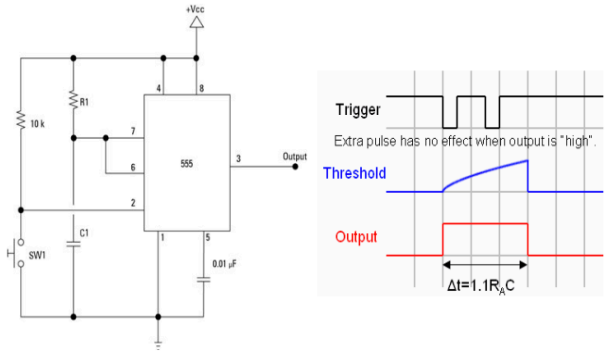


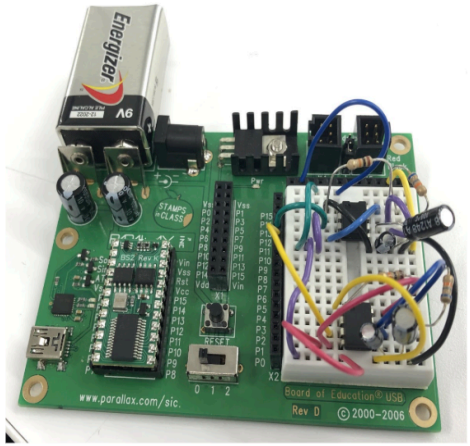
UI Prototype and Wiring

Prototype Build - 555 timer & BOE Board

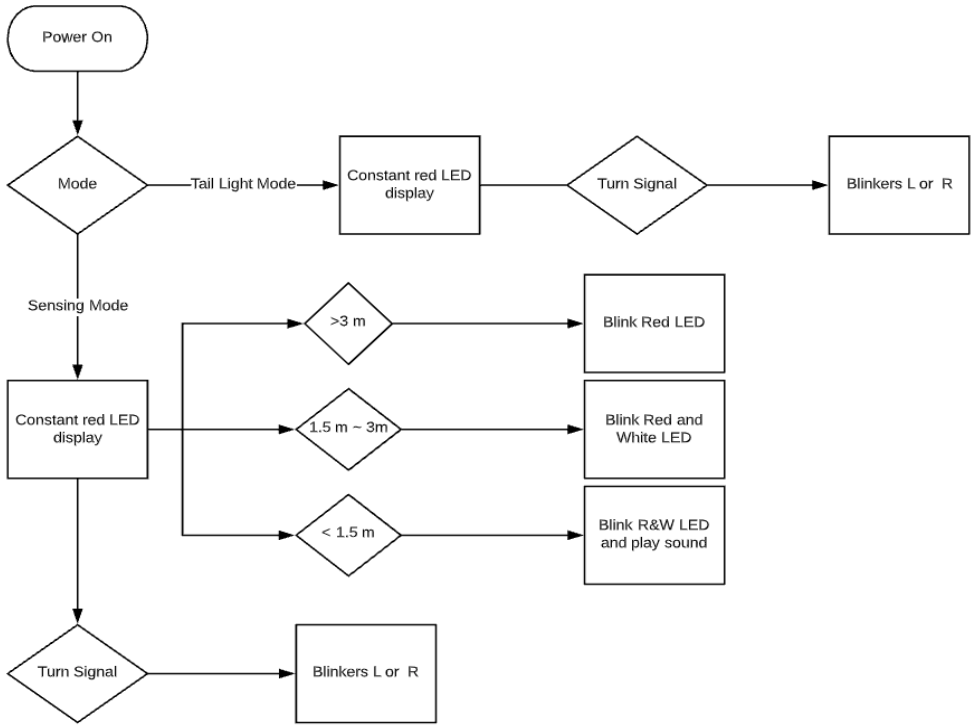
Challenge: Push button pulse not long enough

Solution: 1 shot 555 to delay signal





Final Prototype Build

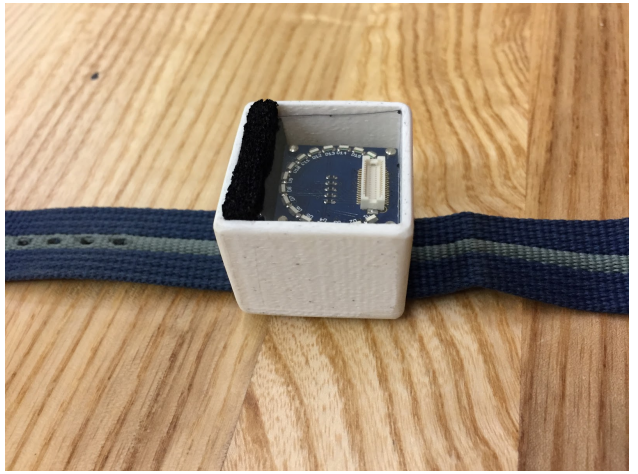


Decision Flow Chart of System

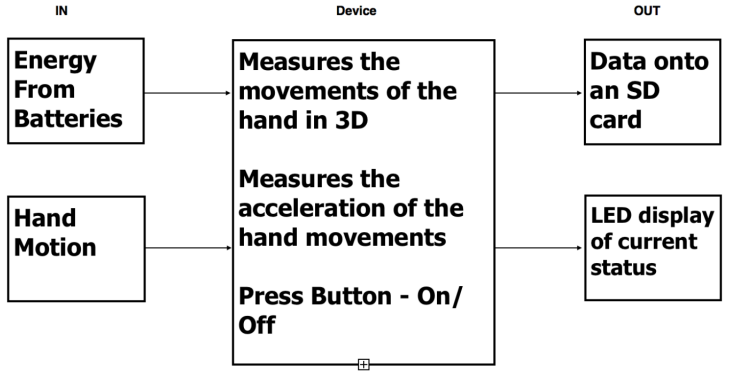
System Mounted on Bike

Hand Motion Monitor for RTS Patients

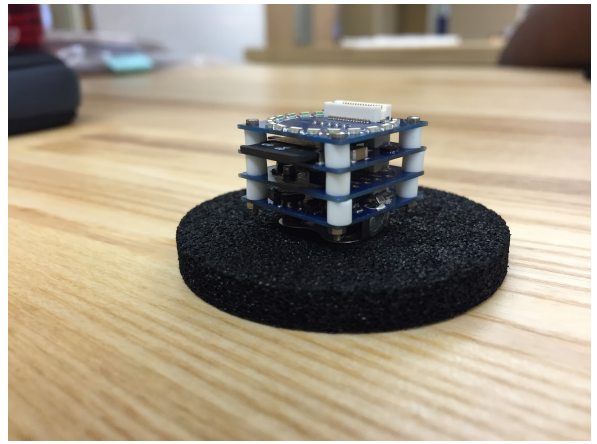
- Project Description:
  - The goal of this project is to create a wearable device that tracks and records the hand motion of RTS (Rett Syndrome) patients for clinical trials
  - Client requirement: Minimum sample rate of 30 Hz, compact, and comfortable for wearing for 12+ hours with matching data storing and power capability.
- Design Process:
  - This project was done back when smart watches such as *Apple Watch* has yet been released not in the market, so the concept of a wearable smart device similar to watch was original.
  - Functional Analysis and black box diagram was constructed in order to further solidify the design.

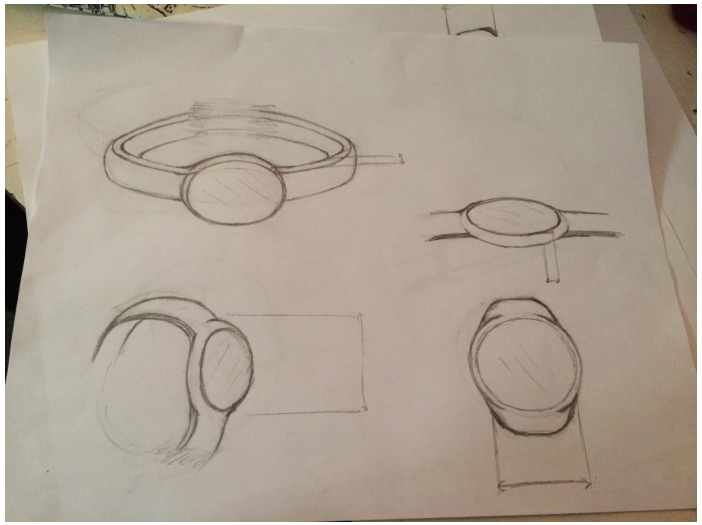


FUNCTION	MEANS		
Attachment to wrist and adjustability for girls ages 1-18	"Baseball cap" straps (watch straps)	Magnetic slap bracelet	Unibody plastic casing
Physical/drool damage resistance for usage life of +/- 5 years	Water-resistance spray for electrical components	thin plastic covering	~
Accurate measurement of the change in position of hand movements	Accelerometer and gyroscope	~	~
Display device status	LED Lights	Mini LED Display screen	~
Data transfer and storage	microSD card	microSD + microUSB cable	



- **Prototyping**:
  - Hardware and software: Programmable microcontroller ”TinyDuino” was used for this project. Similar to Arduino, the TinyDuino runs on C/C++ but is significantly smaller therefore more suitable for a portable wrist monitor. Additionally, the TinyDuino has shields, or add-on’s, that could be stacked up vertically and paired with the microcontroller. The shields that was integrated in this project was the battery module, the accelerometer module, the SD card module, and the LED light module for battery indication and user interface.
  - Housing: The housing for the internal electrical components is a 3-D printed case attached onto a watch strap.
  - Results: Final prototype yielded a sample rate of 125 Hz, way beyond customer requirement. This provides a 8ms read rate, which in return can provide a even more detailed and accurate data. The data is stored in a SD card with the X-Y-X relative positions over certain period of time that can be further investigated by medical researchers.





READINGS.TXT

```
139,159,-172
173,77,-112
167,188,-135
163,168,-133
168,142,-88
188,86,-120
152,144,-172
157,84,-166
139,168,-247
171,114,-126
195,138,-117
155,99,-152
159,18,-135
170,44,-126
165,76,-141
181,160,-93
170,176,-100
149,164,-160
147,154,-128
185,174,-59
155,208,-53
165,106,-33
183,76,18
-150,259,-237
346,185,-34
116,185,-157
116,351,14
72,164,-58
117,129,-61
109,112,-115
```

Final_Code_for_EK210_Project | Arduino 1.6.3

Final_Code_for_EK210_Project

```
#include <SD.h>
#include <SPI.h>
#include <Wire.h>
#include <BMA250.h>
#include <FlexiTimer2.h>

BMA250 accel;
File myFile;

boolean initializationFailed = false;

void setup(){
  pinMode(13, OUTPUT);
  Serial.begin(115200);
  Wire.begin();
  accel.begin(BMA250_range_2g, BMA250_update_time_8ms);//Sets up the BMA250 acceleror
  while (!Serial) {
   ; // wait for serial port to connect. Needed for Leonardo only
  }


  Serial.print("Initializing SD card...");
  // On the Ethernet Shield, CS is pin 4. It's set as an output by default.
  // Note that even if it's not used as the CS pin, the hardware SS pin
  // (10 on most Arduino boards, 53 on the Mega) must be left as an output
  // or the SD library functions will not work.
   pinMode(10, OUTPUT);

  if (!SD.begin(4)) {
    Serial.println("initialization failed!");
    initializationFailed = true;
    return;
    }
  Serial.println("initialization done.");

  // open the file. note that only one file can be open at a time,
  // so you have to close this one before opening another.
  while (SD.exists("Readings.txt")) SD.remove("Readings.txt");
```

Done Saving.

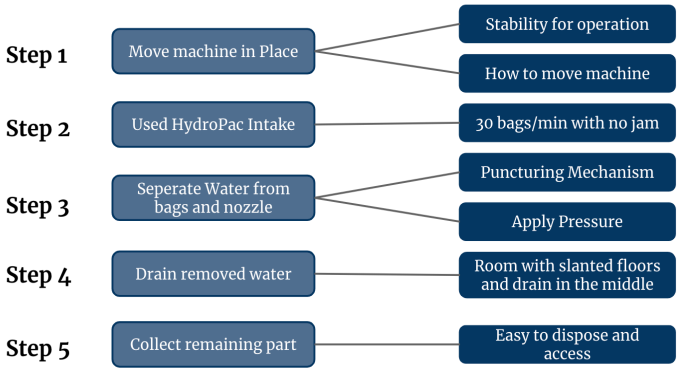
```
The sketch name had to be modified. Sketch names can only consist
of ASCII characters and numbers (but cannot start with a number).
They should also be less than 64 characters long.
```

Arduino Pro or Pro Mini, ATmega328 (3.3V, 8 MHz) on /dev/cu.usbserial-A102SX4K

HydroPac Waste Disposal System

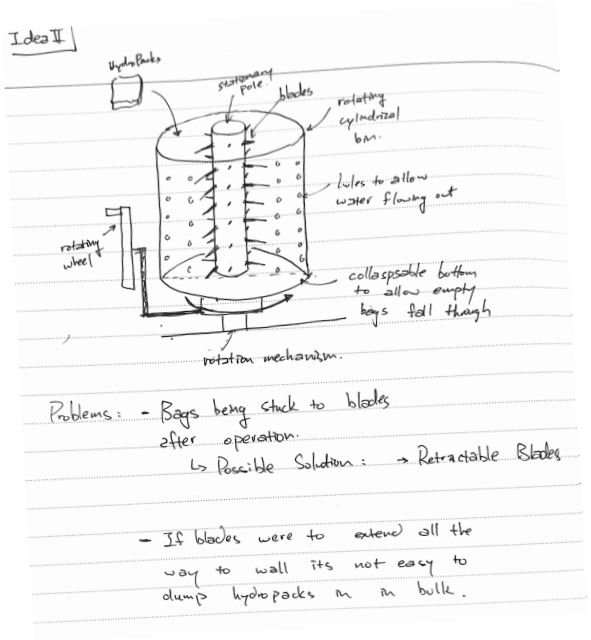
- Project Description:
  - Designed a device to reduce the weight and volume of used HydroPacs for client, Boston University Animal Center.
  - HydroPacs are water pouches for feeding animals such as rats or primates. A plastic valve punctures the water pouch and serves as the nozzle for animals to drink from. Once used, the HydroPacs must be disposed to prevent cross contamination.
  - Disposed HydroPacs often contain leftover water so the aim of this project is to eliminate the water from the HydroPacs using a device that can handle large quantities of HydroPacs and separate the water in an efficient manner.

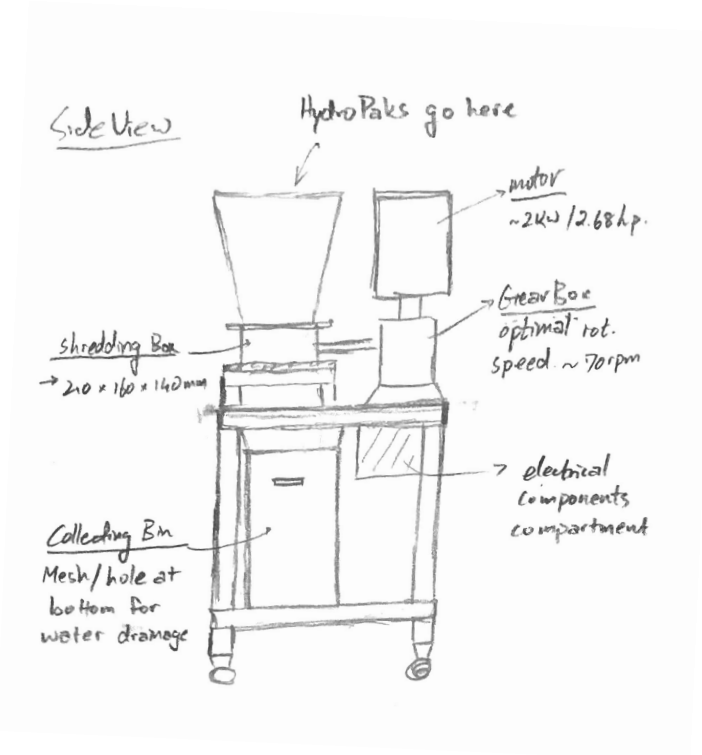
- Brainstorming ideas: A functional decomposition was performed followed by a morph chart analysis.



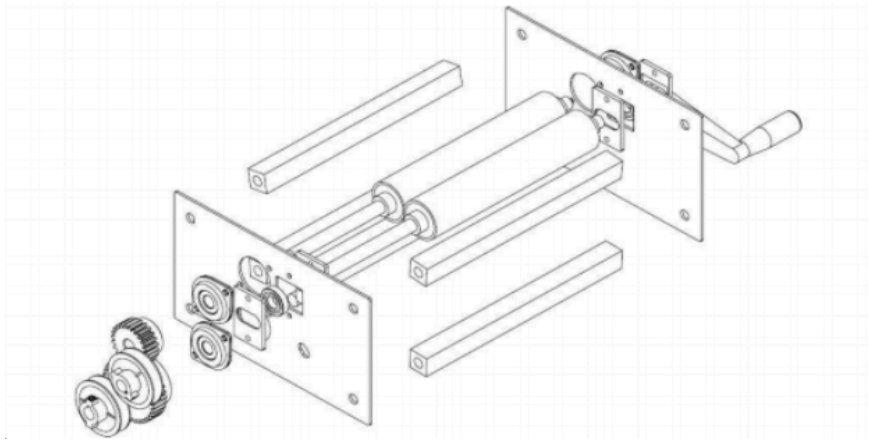
Method/Function	A	B	C	D
Move machine in place	Wheels	Manually lift	Rails	-
Intake Bags without jamming	Hopper	Robotic Arms	Conveyer Belt	Slanted Platform
Separate Water from Bag	Tear with knife	Compression	Shredder	Rollers with spikes
Drain water	Funnel down to drain	Naturally fall down	Piping	Separate Compartment
Separate water from plastic waste	Mesh bin with holes	Mesh Tray	Slanted mesh plate with trash bag	Fishing Net
Collect waste	Door	Bin	Sliding Mechanism	-

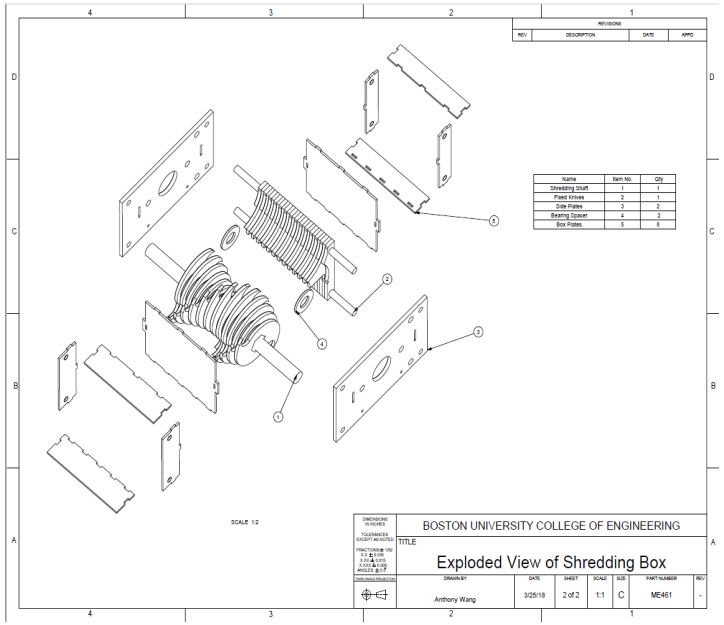
- Preliminary Designs:
  - Utilize the concept of a traditional laundry machine and install blades inside, so that HydroPacs gets punctured and water can flow out of cylinder
  - A shredding device that destroys both the thin film water pouch as well as the polypropylene valve and collects the shredded parts through a filter.
  - A roller that compresses the HydroPac so that water squeezes out of the HydroPac due to high pressure under roller.
- Remarks: Initially the roller design was favored but the surface of the roller provides little friction to grip on to HydroPac and therefore the shredder design was chosen.
- A feasibility analysis was also performed in order to calculate the required motor power to shred the HydroPacs.

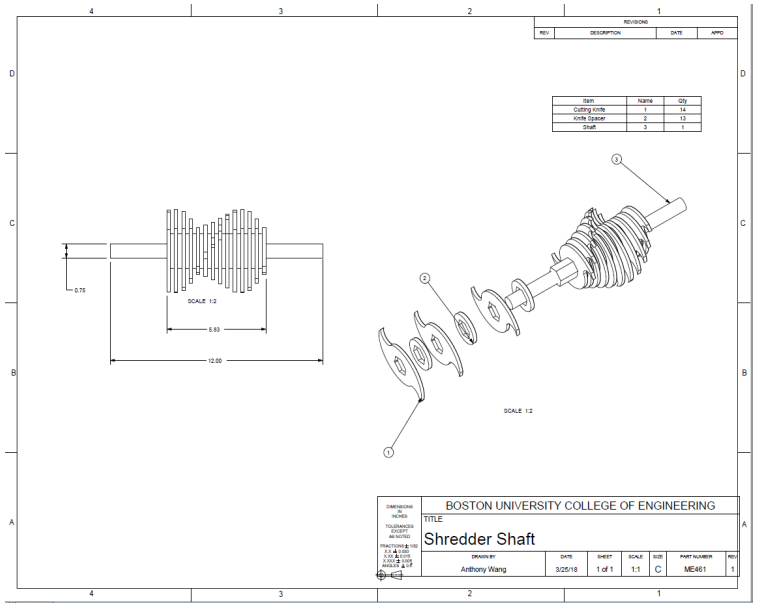




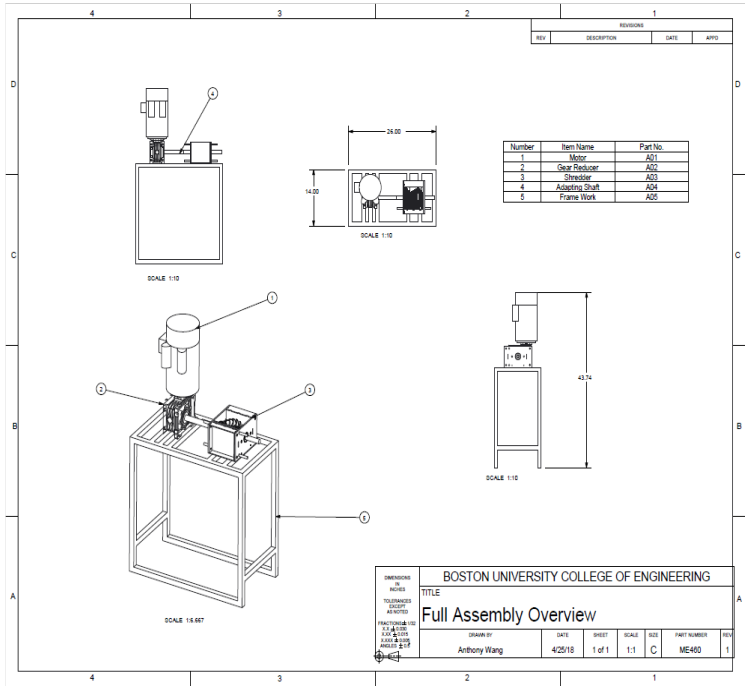
Valve Length	in	1	Area	in^2	0.075
Valve Thickness	in	0.15	Area	ft^2	0.000520833
Yield Strength of Plastic[PP]	MPa	40	Cutting Edge Length	in	0.5
Yield Strength of Plastic	Psi	5800	Cutting Radius	in	2.36
Force to shear one bag	lb	435	Half of Cutting Radius	in	1.18
Torque	lb-ft	128.325	No. of teeth	-	3
RPM		52	HydroPac Length	in	5.6
Motor Power (hp)	hp	1.905817	Disposed HydroPac Height	in	0.5
Design Factor	-	1.5			





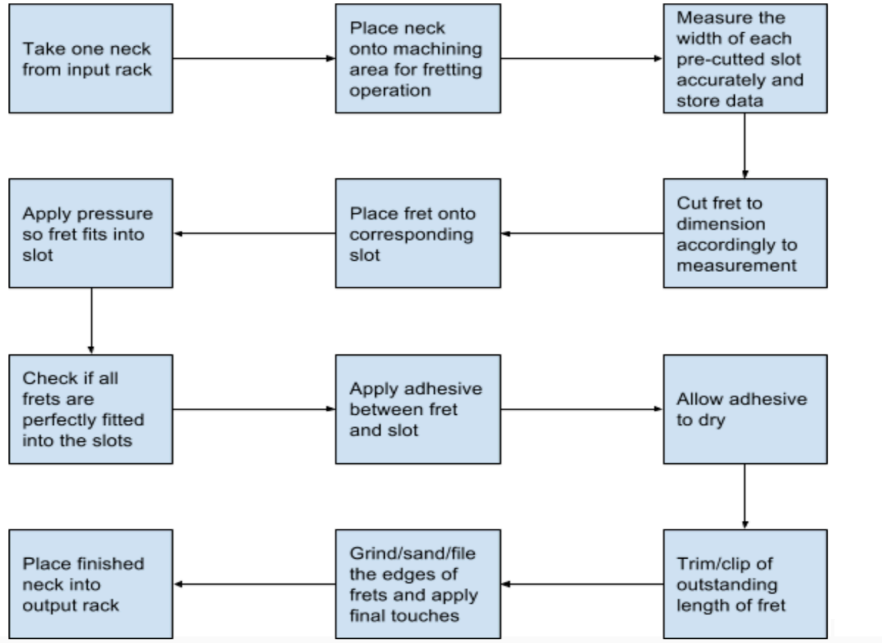


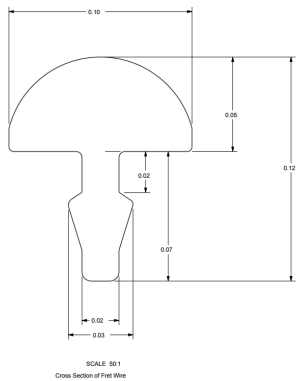
- CAD Models, Finished Product, and Results
  - Videos of the device demo can be found at: https://youtu.be/RURnBDdWcDk https://youtu.be/i6y99W_L01c
  - Sheet metal was lasercutted, deburred, and assembled
  - The supporting structure was built with 80-20s
- This project was nominated for Societal Impact Award among Senior Design Projects.

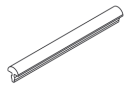


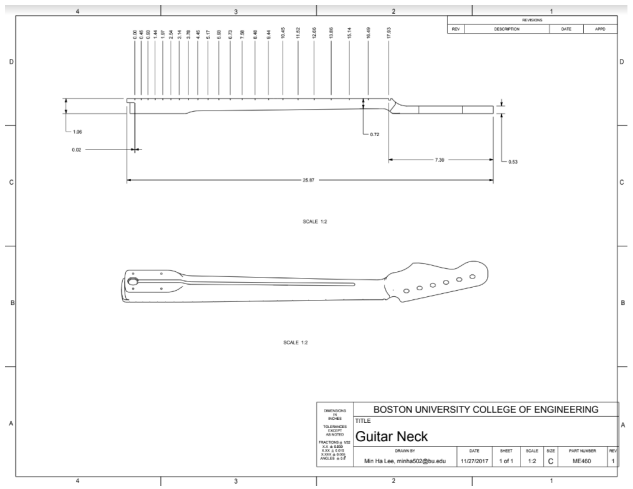
Automated Guitar Fretter

- Project Description:
  - Traditionally, the process of fretting a guitar is done manually by pressing metal frets into pre-slotted guitar necks.
  - The frets are then trimmed down and polished to the precise fit which takes over an hour for the operation.
  - The goal of this project is to automate the process of fretting a guitar and meeting the following requirements:
    - 10 guitar necks per cycle
    - 5 seconds per fret, 21 frets per neck
    - 60 seconds for trimming
- Functional Decomposition and Morph Chart Analysis during the brainstorming phase.



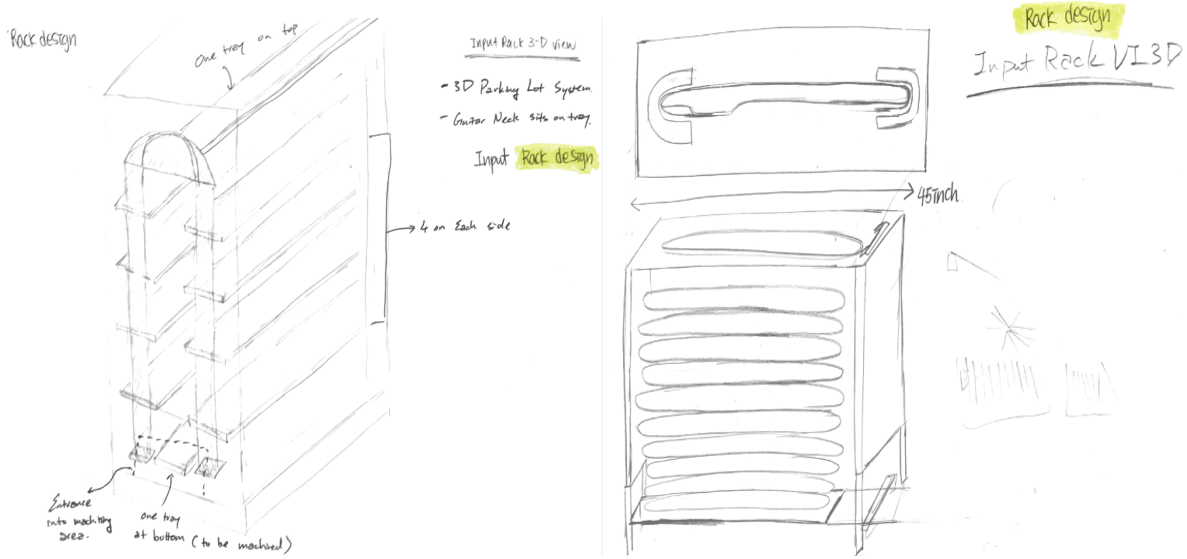


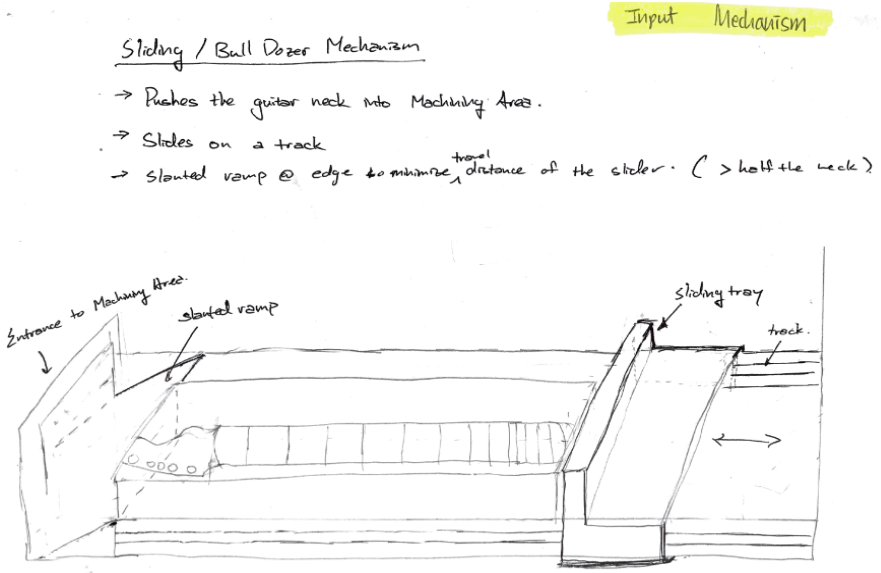


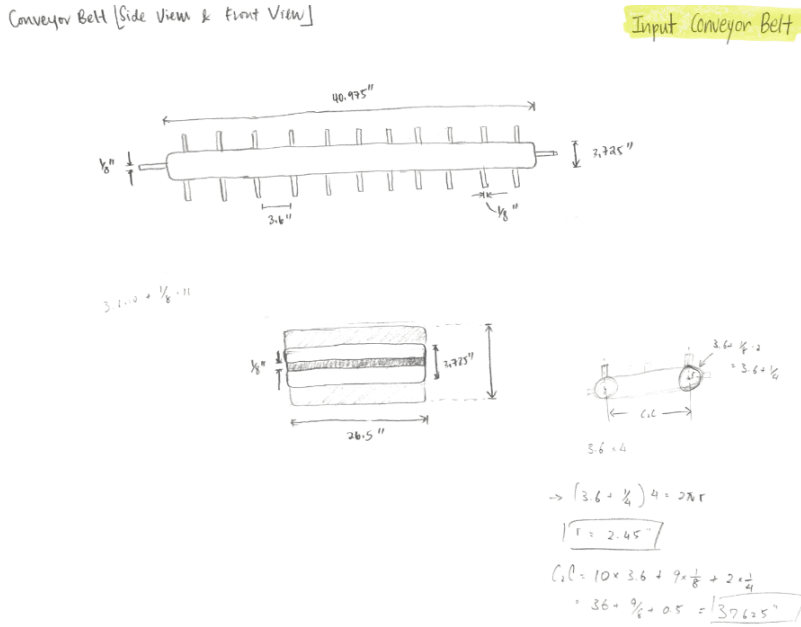


Method/Function	A	B	C
Take neck from rack	Robotic arms	Slanted rack	Linear drive
Place in machining area	Conveyor belt	Robotic arms	Friction belt
Measure slot width	Laser sensor	Limit Switch	Imaging
Cut fret to dimension	Press fret into fixed blade	Laser cut	Rotary tool
Place fret on slot	Robotic arms	Move neck under fret	-
Fit fret into slot	Pneumatic	Hammer	Roller
Trim extra parts	Grinder	Laser cutter	Water cutter
Edge finish and final touches	Fine sand paper	Rotating steel wool/brush	-
Move finished neck to rack	Robotic arms	Conveyor belt	Pushing motion

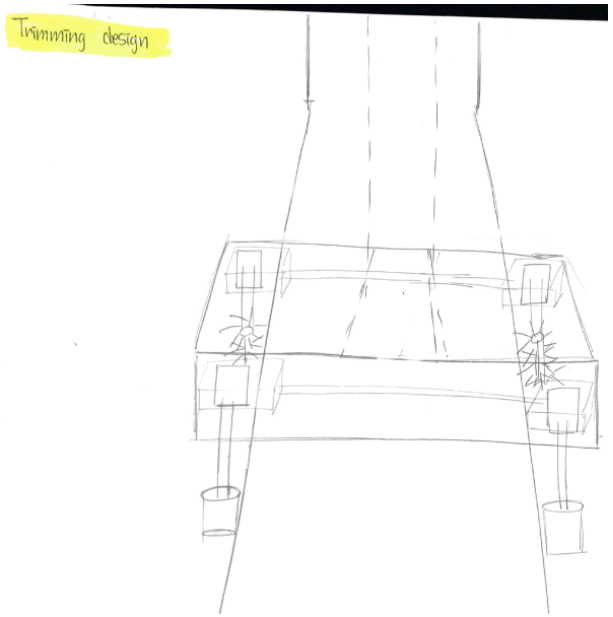
- Preliminary Designs: Since this is a full-scale assembly line, different stages were isolated first so that the design of each stage can be inspected in detail
- Input/output Rack:
  - Automatically feeds in pre-slotted guitar necks and queue the remaining necks.
  - Initial designs stack the neck vertically upwards but was later changed into a horizontal conveyer belt for simplicity and ease for inspection.

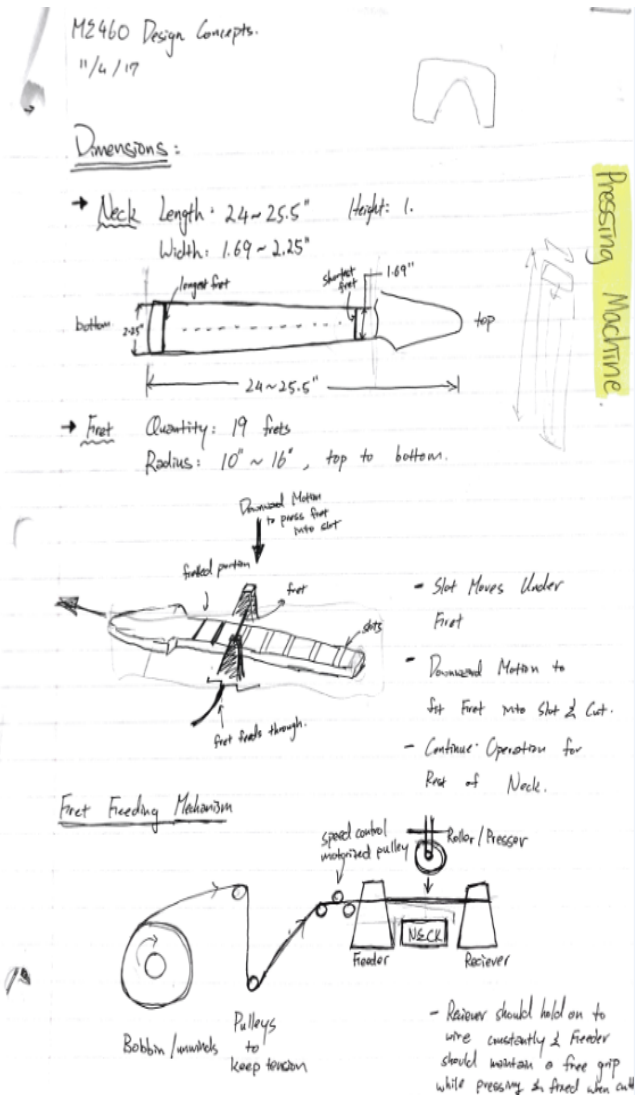


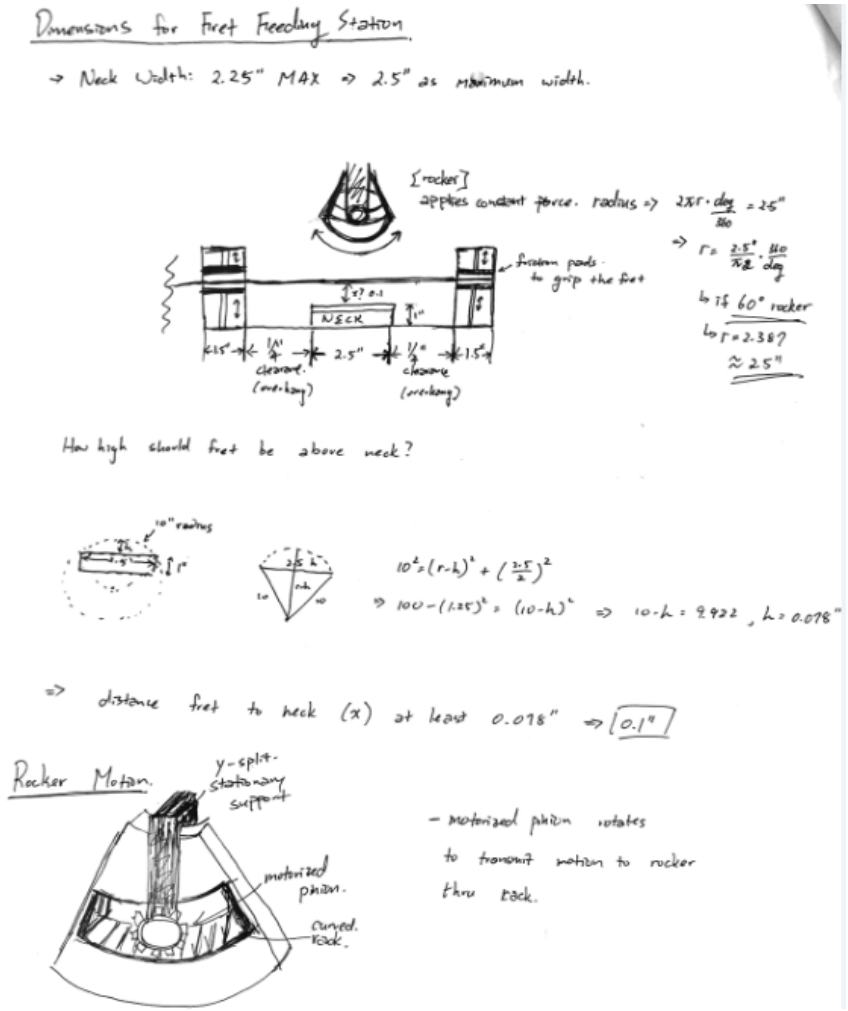




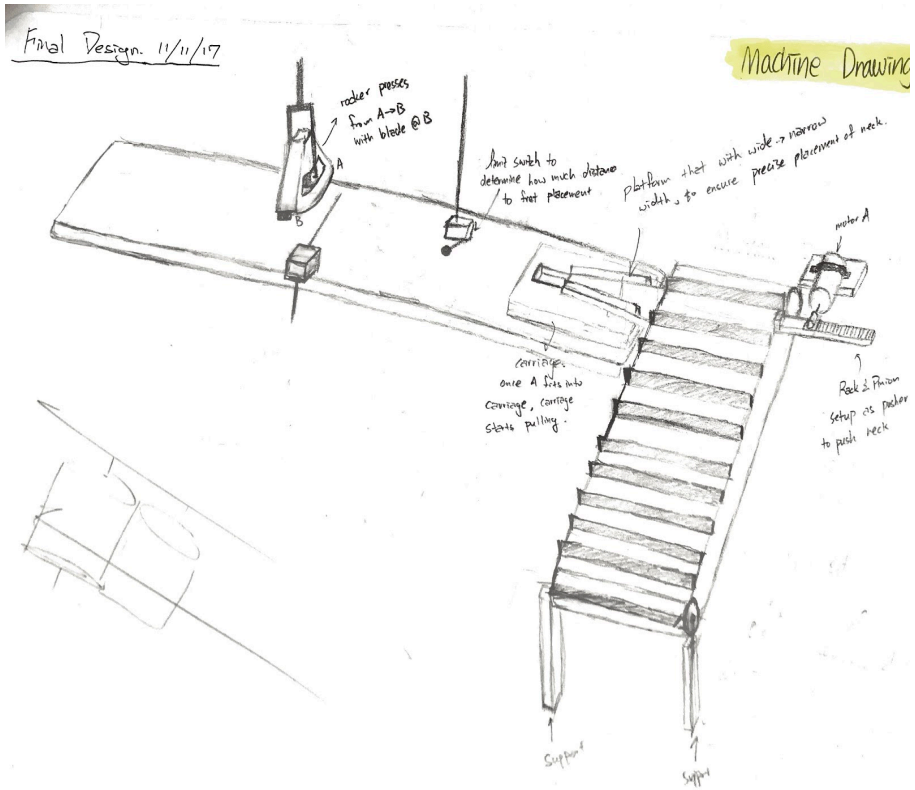
- Fret Feeding System: Ability to keep the tension of the metal fret and feed it precisely to the desired length
- Fret Pressing: Enough force to press the metal fret into the slot without damaging the neck by applying too much pressure.
- Fret Trimming: Adjustable to the varying width of the guitar neck in the axial direction. Able to cut/grind off excess fret and round the edges.

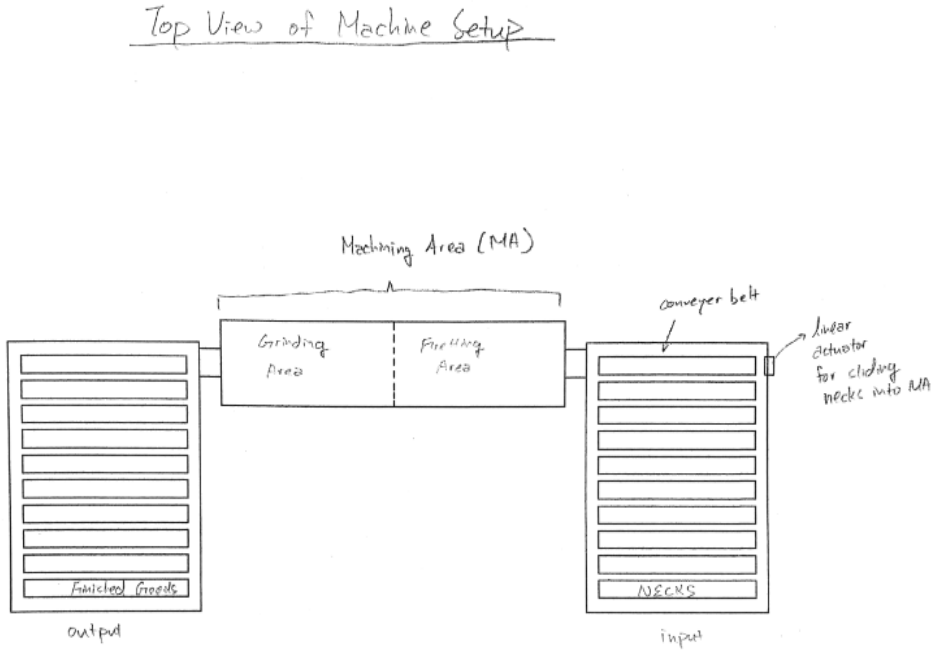




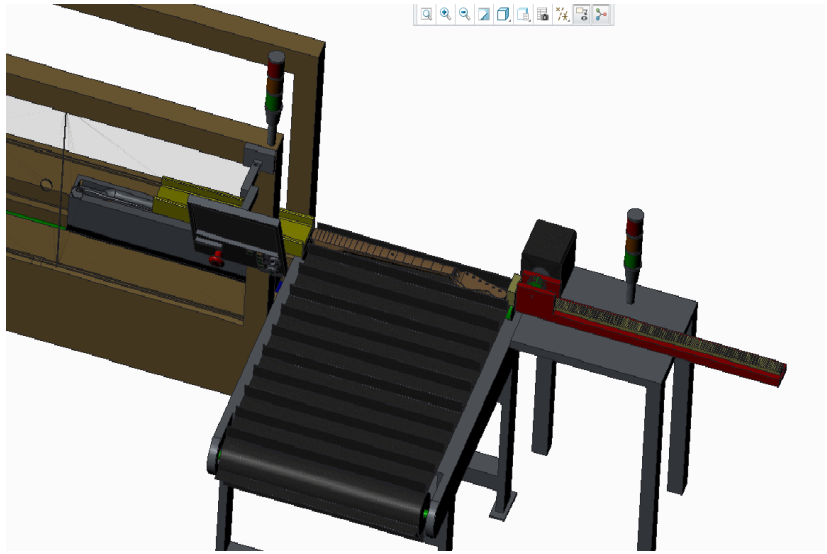


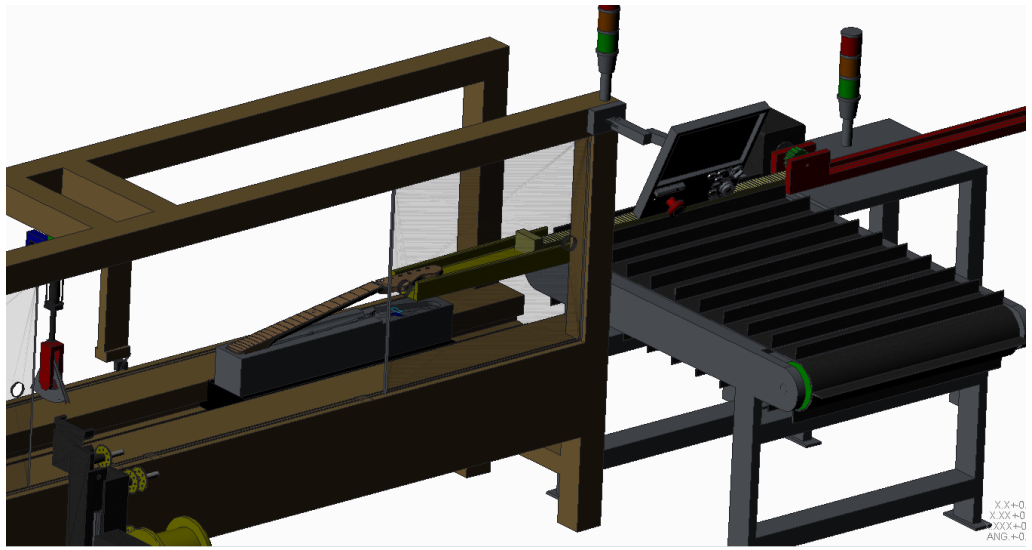
- Putting it all together: After isolating the different operations, a final sketch was put together. The figure on the left does not display to output rack. With the design in mind, the CAD model was also sketched out.

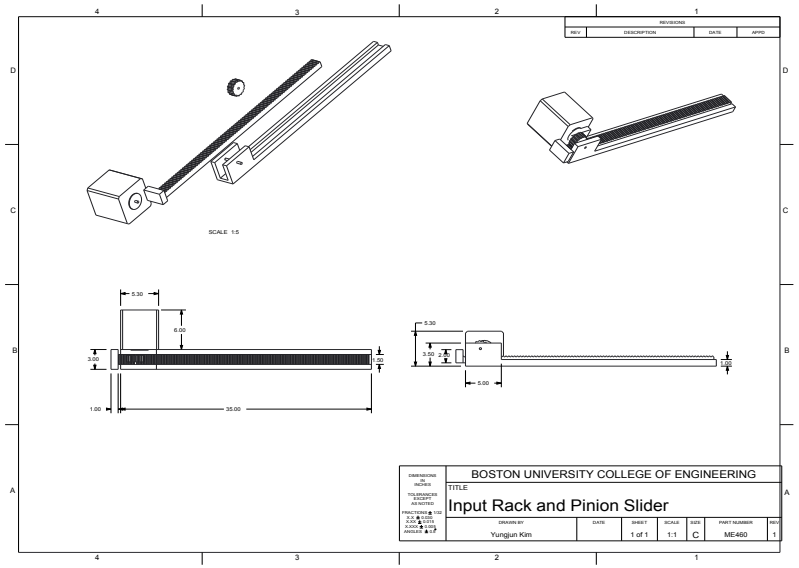


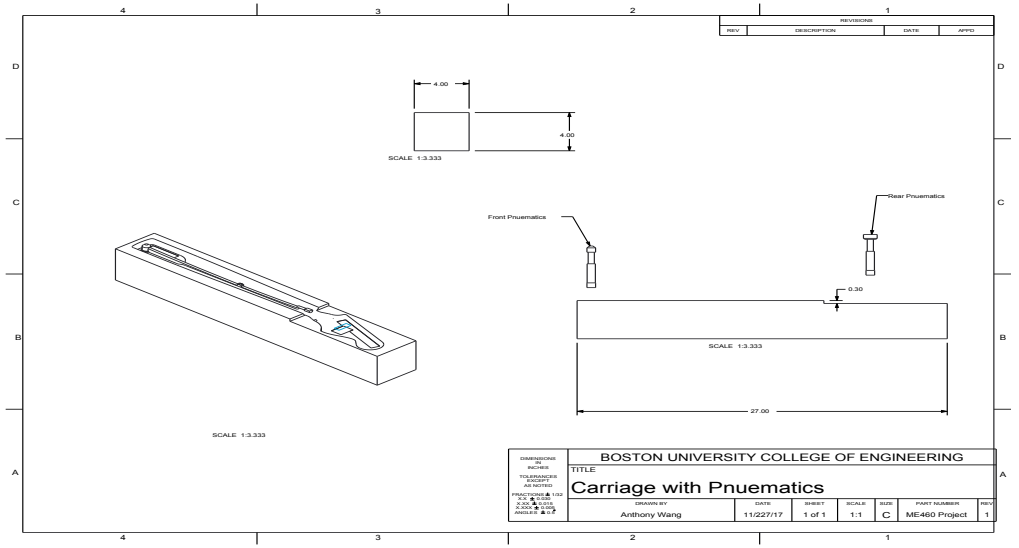


- Input Rack: Pushes the guitar neck into the machine and gets transported by a carriage that travels along the machine. The Conveyer Belt then transports the following neck and queue for operation.



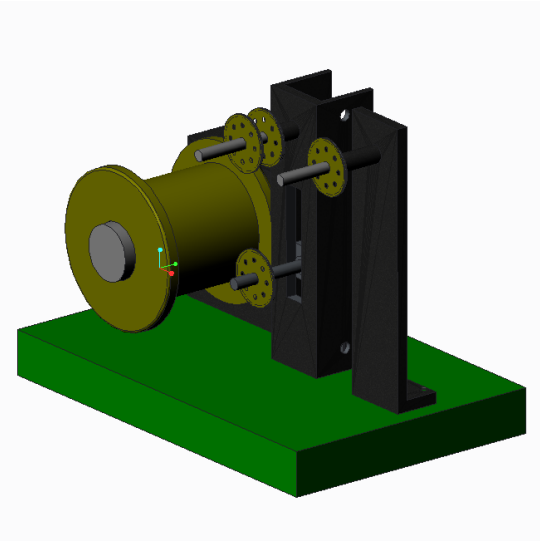


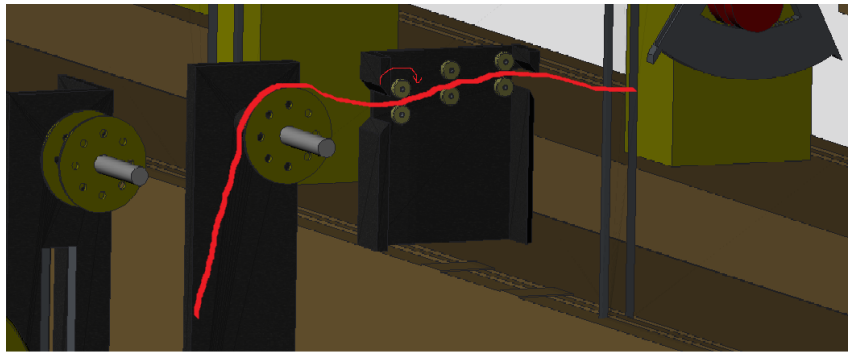


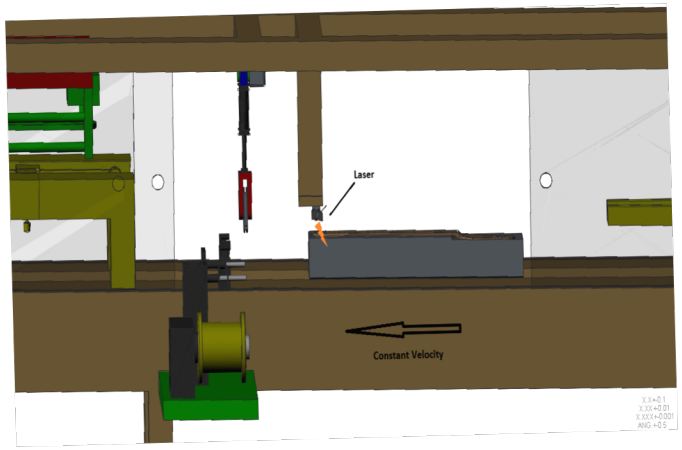


Carriage that is installed into the machine and replaceable for different necks

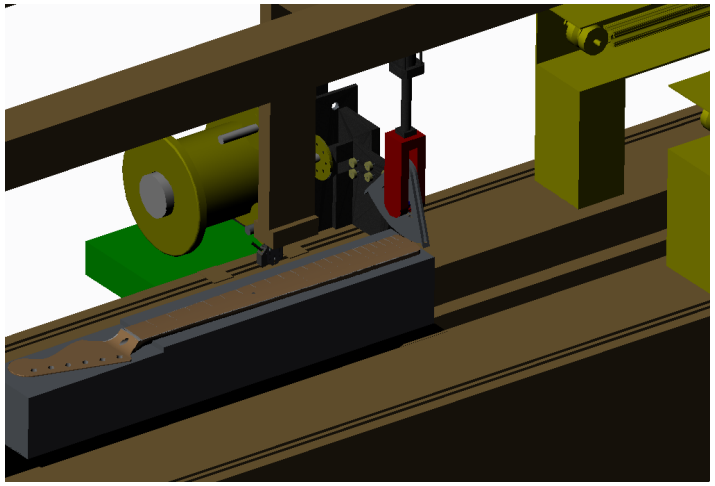
- Fret Feeding System is consist a bobbin, or a roll, of guitar frets. The frets are unwind and fed through a series of pulley that moves up and down to maintain the tension. It hovers the pre-programmed length of fret over the slot of the guitar for the pressing operation.

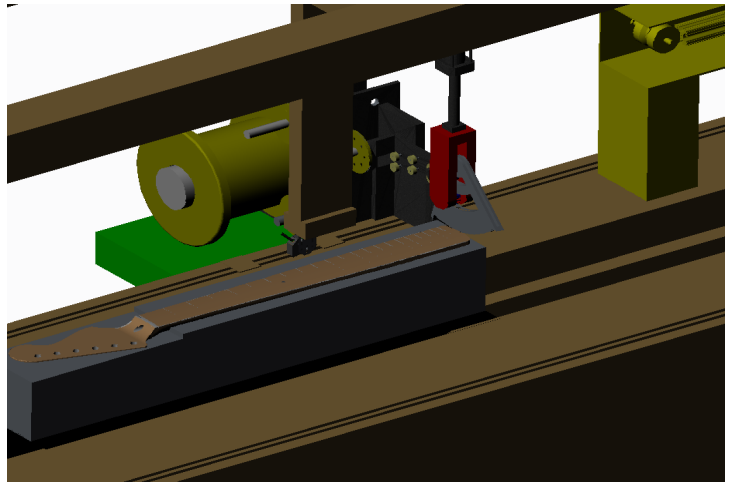




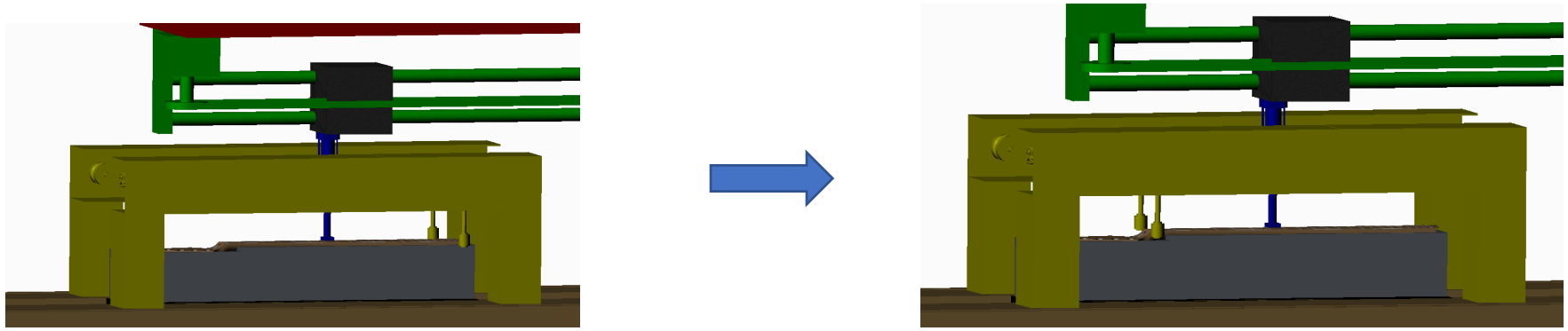


- Pressing system is a pneumatic, constant force press with a rocker actuator. The actuator is lowered until contact and rocks along the contour of the fret to fit the fret into the slot. At the left end, there is a blade to cut off the fret from the fret feeder.

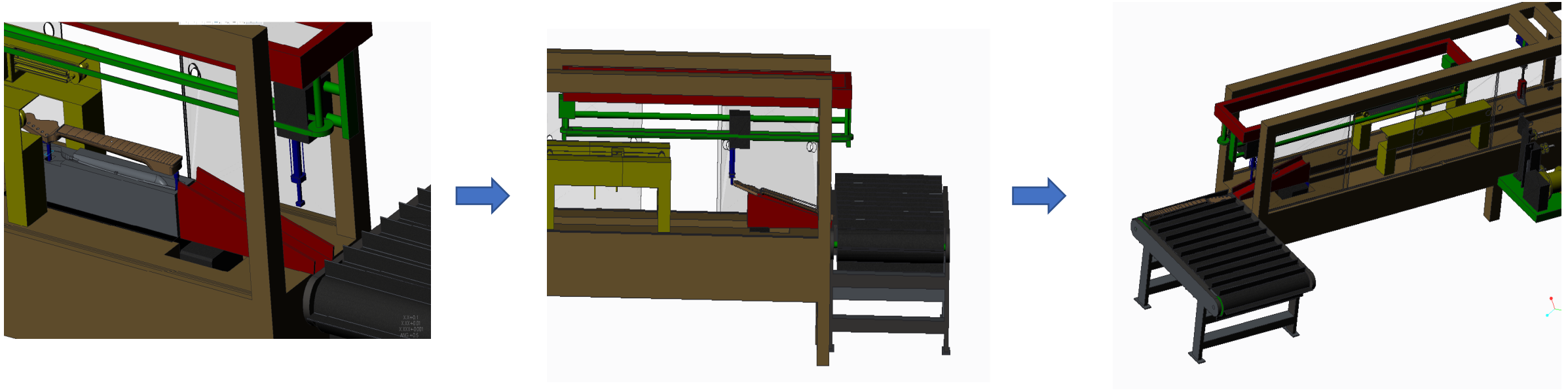




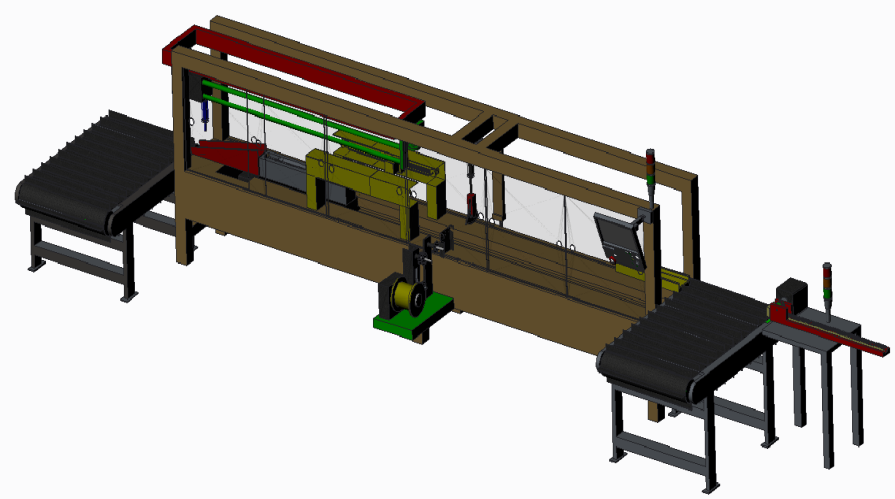
- The trimming operation is done similarly to a car wash machine. A pneumatic press comes down and hold the guitar neck in place. Two pairs of rollers with coarse and fine grinding wheels move along the parameter of the guitar neck to trim off the excess frets.

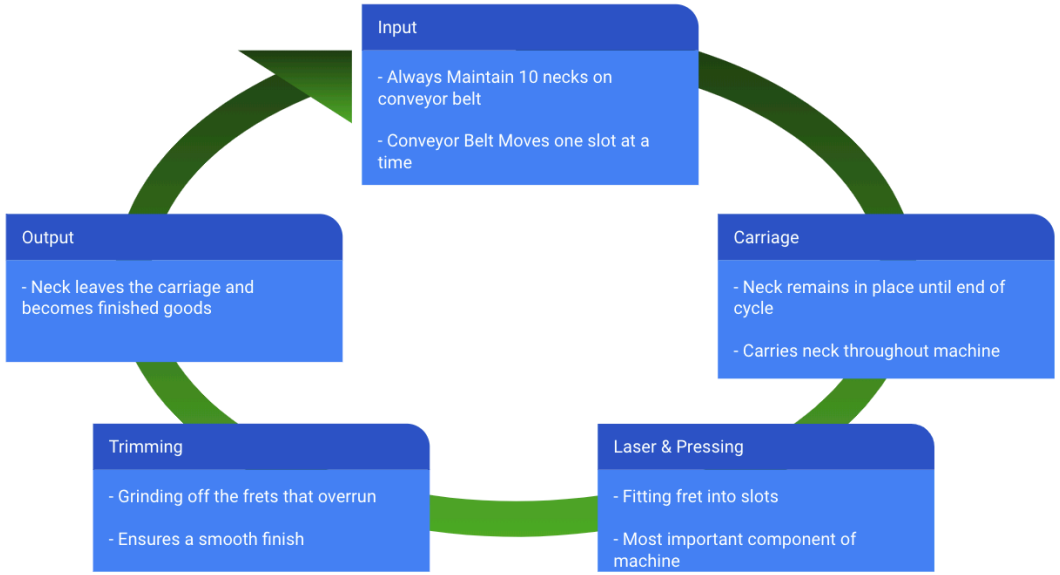


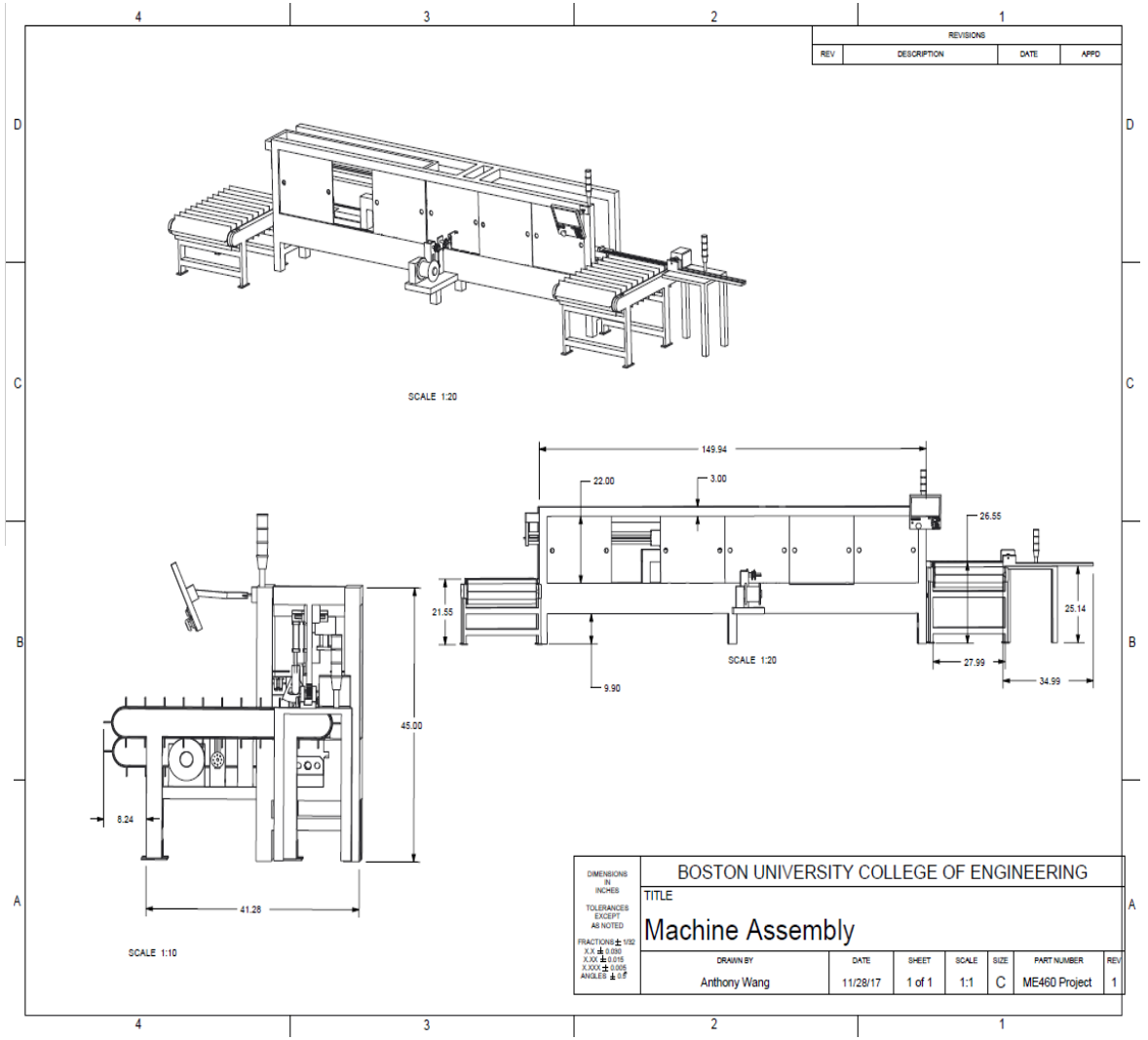
- The carriage then transport the workpiece to the final stage where the built in pneumatics in the carriage elevates the workpiece and the previous pneumatic actuator used for holding down the guitar neck slides the guitar neck onto the ramp which guides it to the final finished goods conveyor belt.



CAD Model of the full assembly line and Operation Cycle

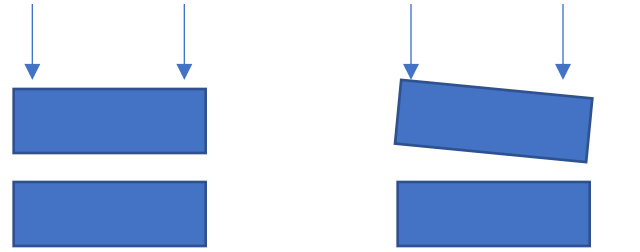






Vertical Displacement Detection System

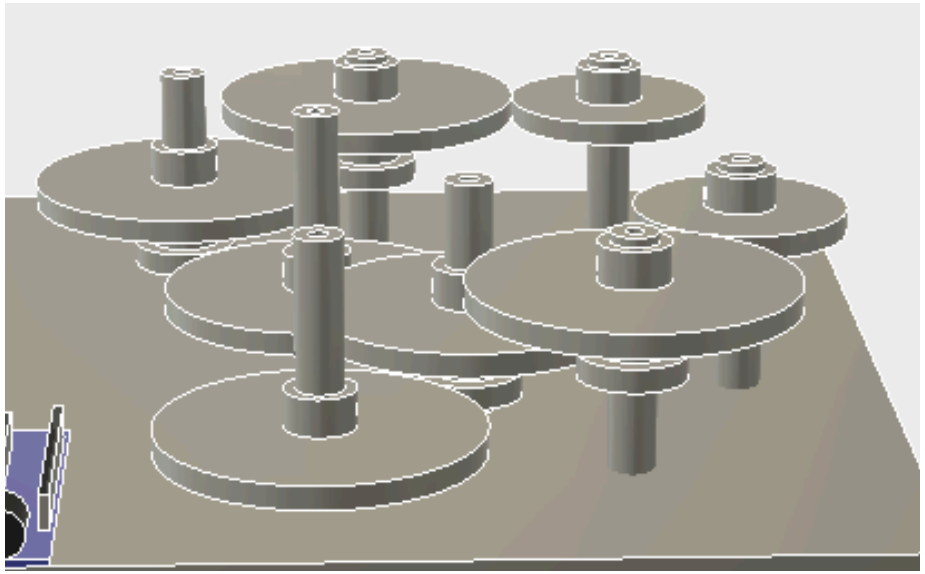
- Project Description:
  - Goal: Create a device that monitors the minor vertical displacement (10 ^ -4 mm) of a pair of rollers that is being used in the operation of pressing round copper wires into flat wires and display the data.
  - The gap in between the rollers is controlled by a pair of worm gears connected to motors.
  - It is essential for the operation that the two worm gears move in the same increments otherwise the difference in elevation forms an uneven surface on the finished product.
- Design Process:
  - Since the rotation of the worm gear directly controls the elevation change, if the rotation of the two gears are equal, the displacement will be the same
  - Design Limitation: the rotations increments are extremely small
  - Morph Chart Analysis
  - Feasibility Analysis indicates that the input rotation needs to be amplified 18x in order to be a detectable signal.

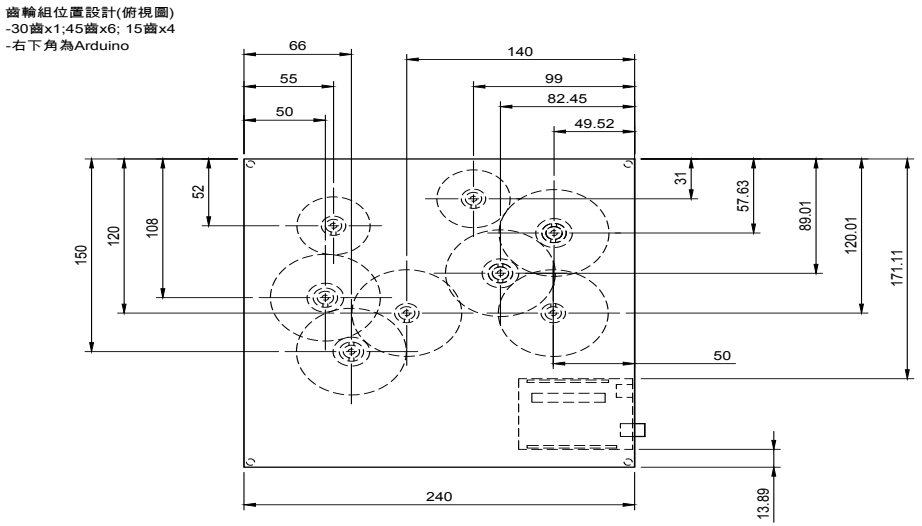


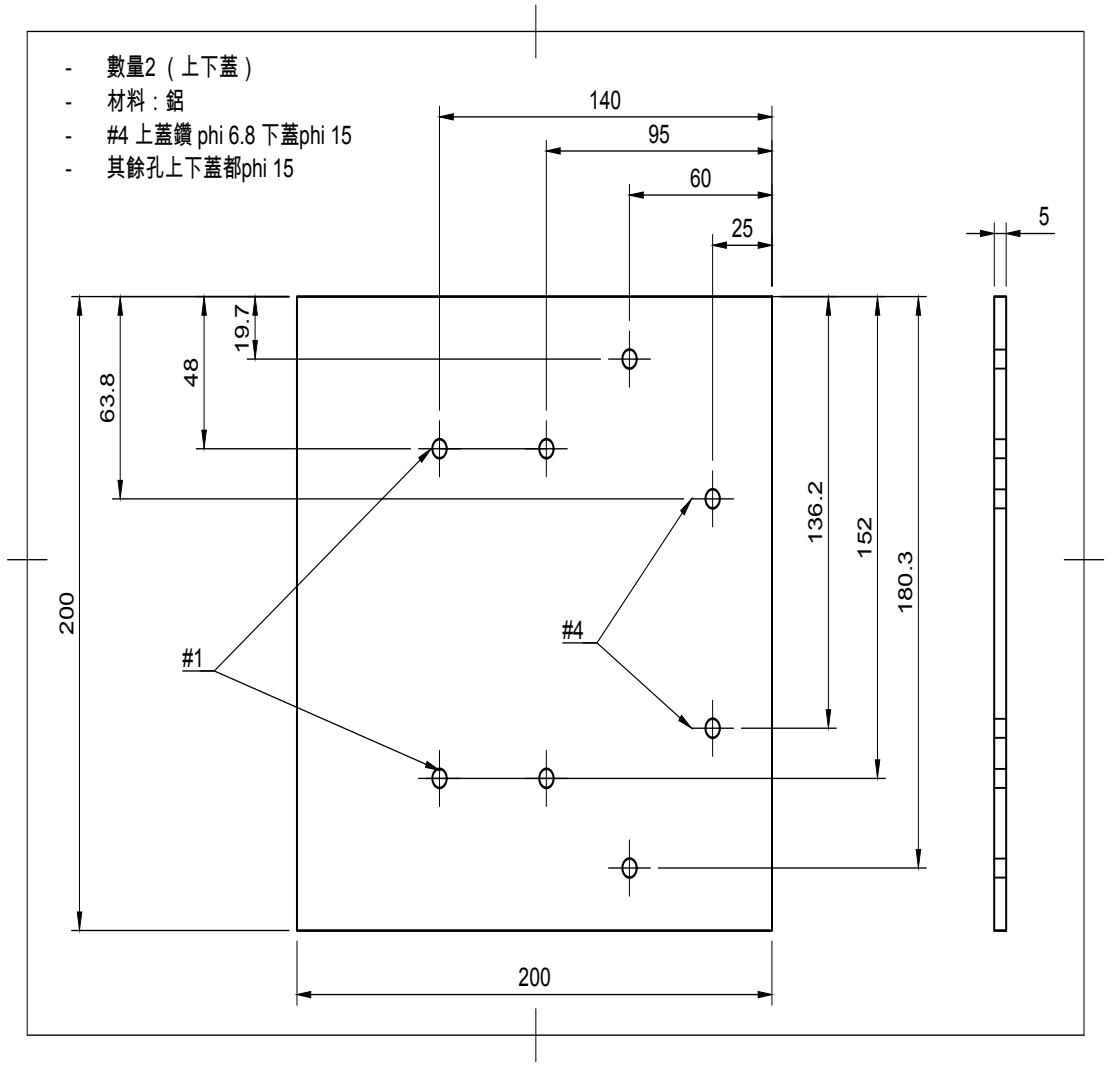
Even Displacement Uneven Displacement

	Method A	Method B
Detect Rotation	Tracks on Disc (Optical Encoder)	Gear Train
Sensor	Laser	Rotary Encoder
Material	Aluminum Sheet	Acrylic Sheet
Control	Traditional Circuits	Microcontroller (Arduino)

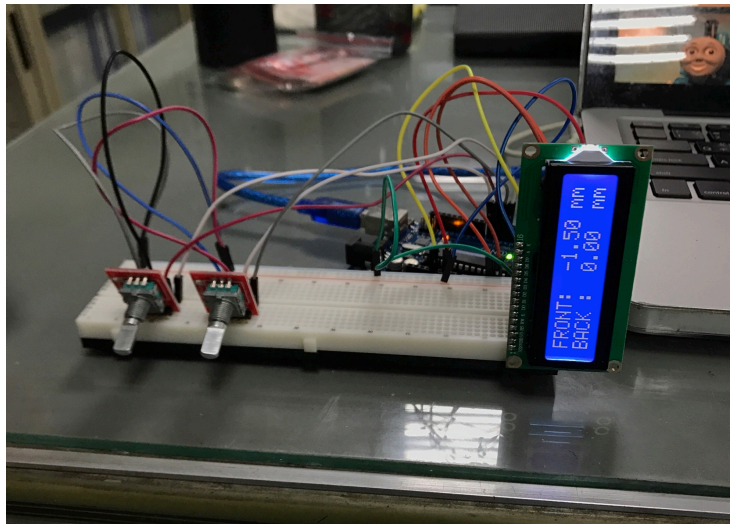
Gear Train Design: After determining the proper gear series, the CAD model was sketched out, as well as the precise location of holes for the shafts.





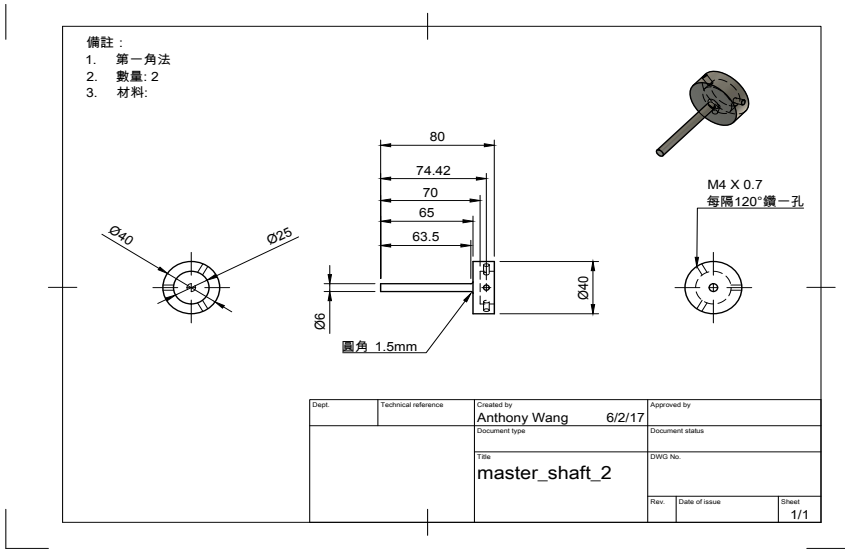


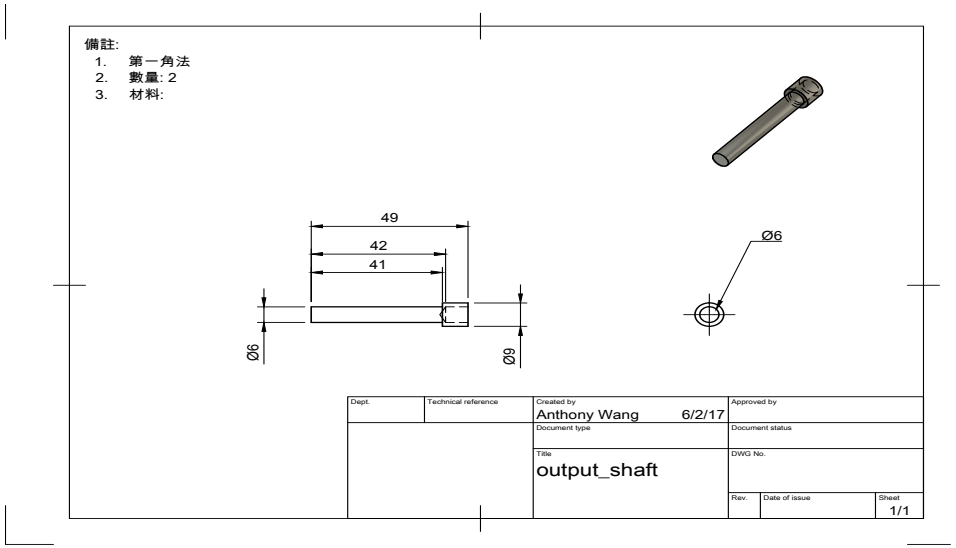
- Software: Coding Arduino and using two encoders



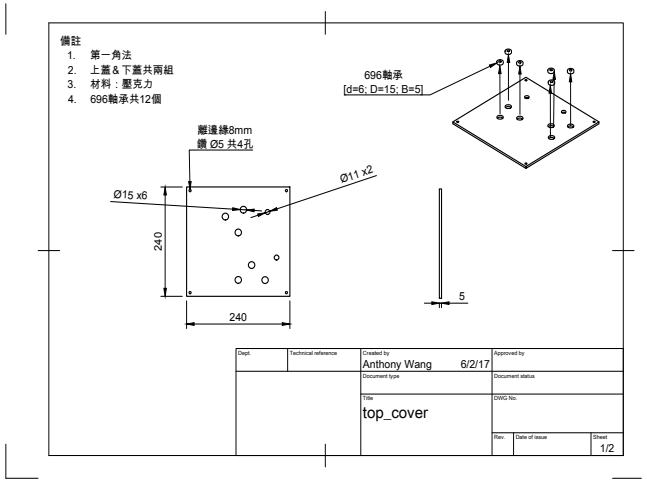

```
Project_test
// include the library code
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
// initialize the library with the numbers of the interface pins
LiquidCrystal_I2C lcd(0x27,16,2); // set the LCD address to 0x27 for a 16 chars and 2 line display
/*********************************************************/

// defines pins numbers
 #define OA1 6
// #define OA2 4
 #define OB1 7
// #define OB2 5
 int counter1 = 0;
// int counter2 = 0;

 float angle1 = 0.000;
 int aState1;
// int aState2;
 int aLastState1;
// int aLastState2;

void setup() {
  pinMode (OA1,INPUT);
  pinMode (OB1,INPUT);
//  pinMode (OA2,INPUT);
//  pinMode (OB2,INPUT);
  lcd.init(); //initialize the lcd
  lcd.backlight(); //open the backlight
  aLastState1 = digitalRead(OA1);
 // aLastState2 = digitalRead(OA2);
  lcd.setCursor(0,0);
  lcd.print("      XXXXXX      ");
  lcd.setCursor(0,1);
  lcd.print("      XXXXXX      ");
  delay(1500);
  lcd.clear();
  lcd.print("FNT:          mm");
 /* lcd.setCursor(0,0);
  lcd.print("      SYSTEM      ");
  lcd.setCursor(0,1);
```

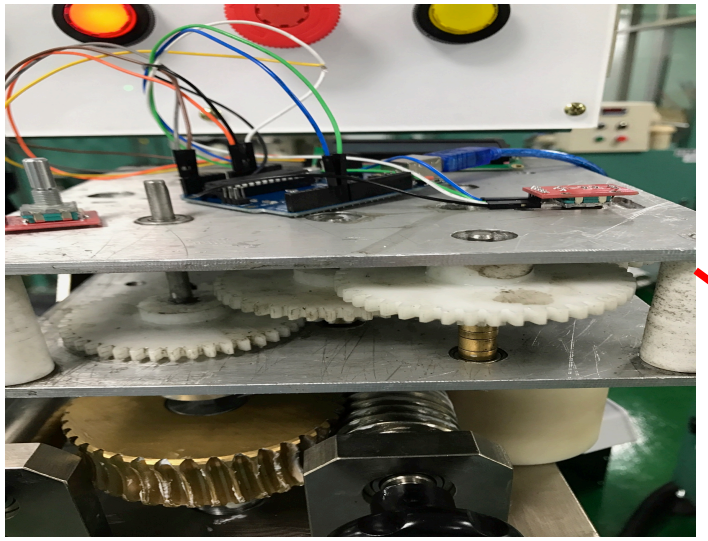
- Hardware: Designing the adapters for input shaft and output to encoder.

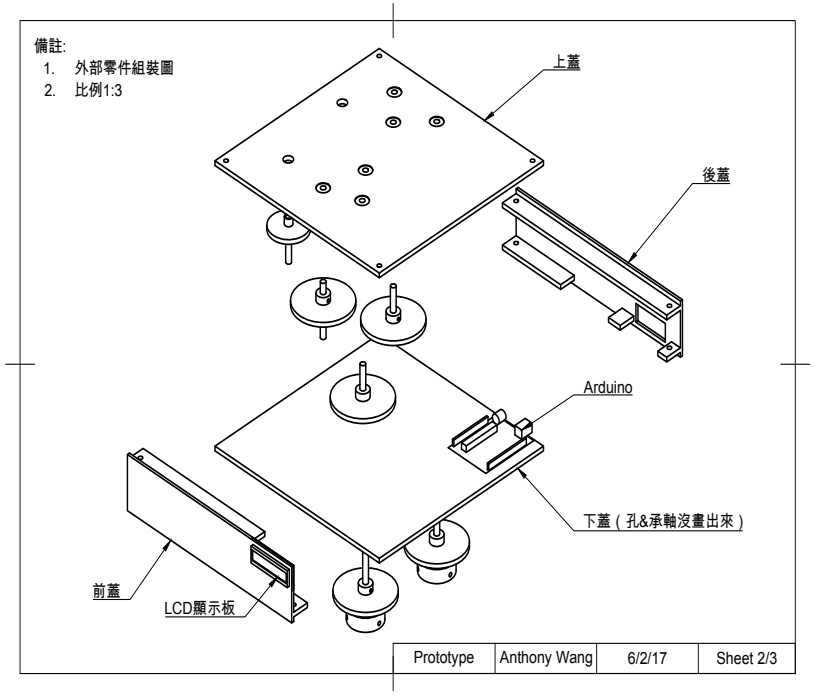


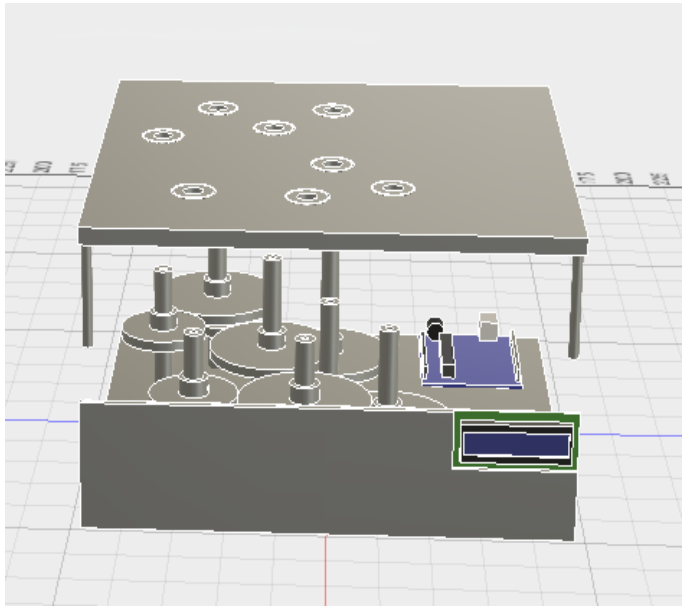


- Putting it all together



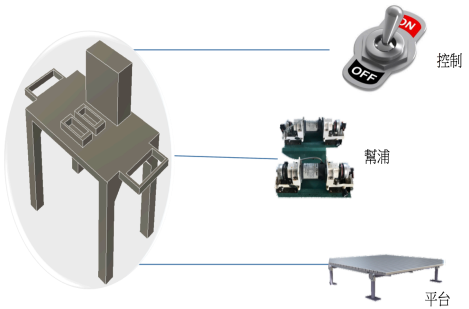


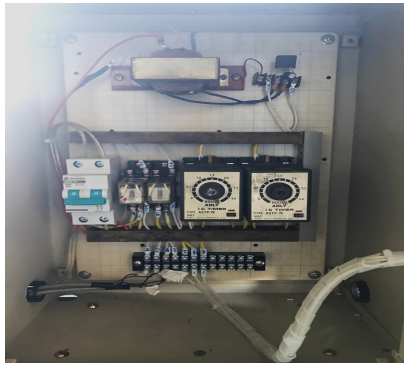




Copper Wire Coating Box Cleaning System

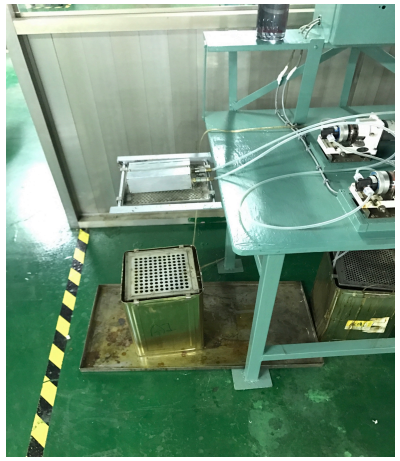
- Project Description:
  - During the manufacturing process of copper wire, a very essential step is to coat the wire with insulation paint which replaces the physical insulation covers in traditional wires. The painting process is done by running the wire through a coating box which is prefilled with chemicals, and the box requires thorough cleaning.
  - The goal of this project is to implement a semi-automated cleaning system that allows multiple boxes to be cleaned during the operation.
- Design:
  - There are existing input and output openings on the boxes that allow both coating materials as well as cleaning chemicals to cycle within the boxes. The openings can be connected to pumps to transport fluid in and out of the boxes.
  - A mesh that allows fluid to drain out of the boxes after cleaning.
  - Implement timer so operators can leave the system running on its own.
  - The cleaning chemical is highly corrosive so the material of the platform must be corrosion resisted.





Internal Wiring

Complete Build



Coating Box Cleaning in Operation

Additional Projects

- Aluminum Pen:
  - ➢ Turned on lathe from solid aluminum cylinder stock
  - ➢ Body of the pen are two pieces, machined to fit and burred out the connecting line
  - ➢ Lightweight, compact, durable.

- Fidget Spinner: Replica of the *Typhoon Fidget Spinner*
- Made with Conventional Milling